



Cardiac arrhythmia & Management

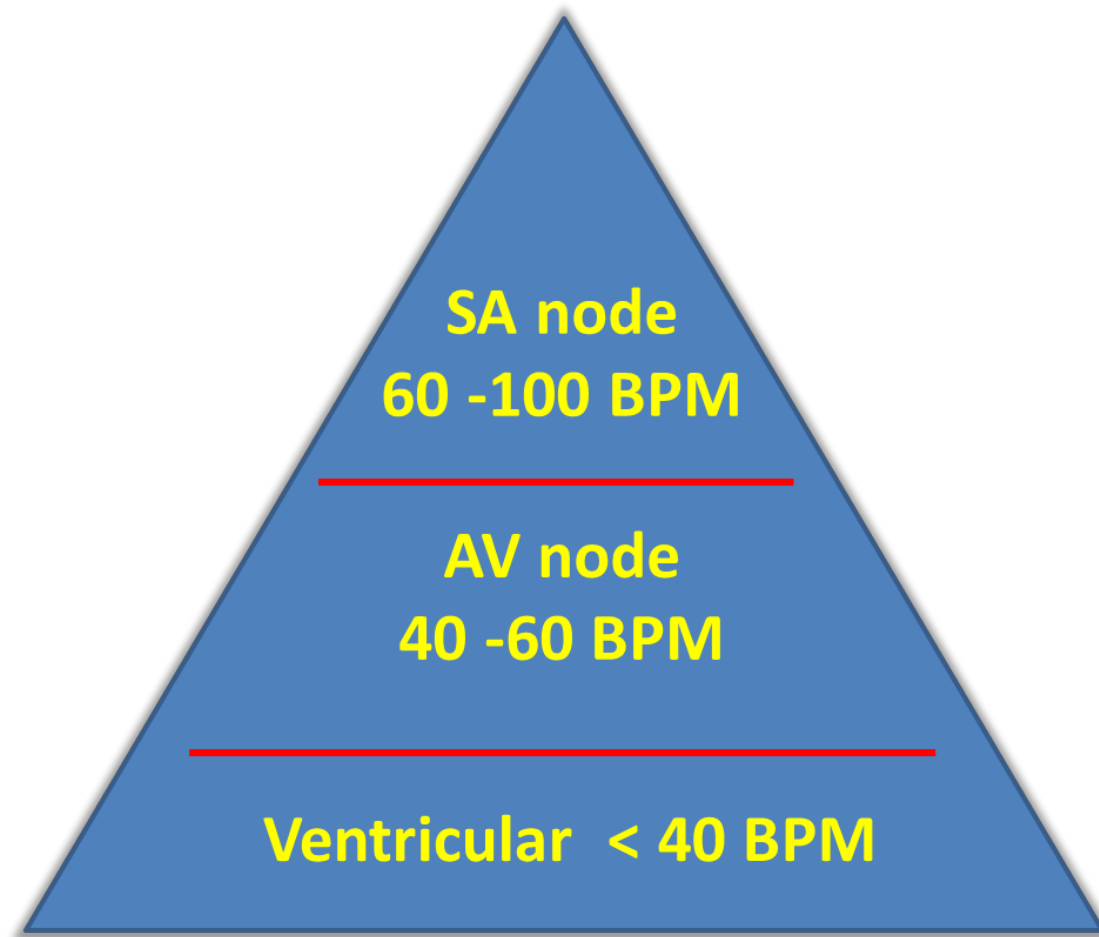
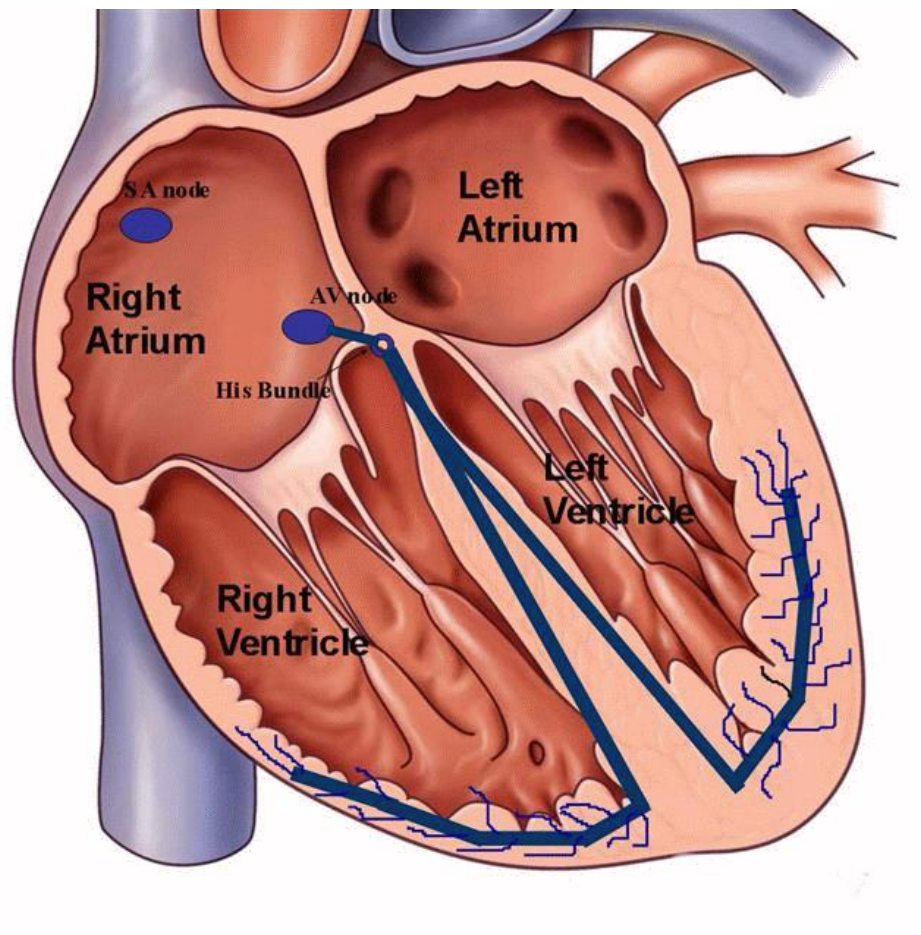
**Pattanaporn Supina
CCU Siriraj.**



- สามารถอ่านและแปลผลคลื่นไฟฟ้าหัวใจ arrhythmia ได้ถูกต้อง
- สามารถให้การพยาบาลผู้ป่วยที่มีภาวะ arrhythmia ได้ถูกต้อง

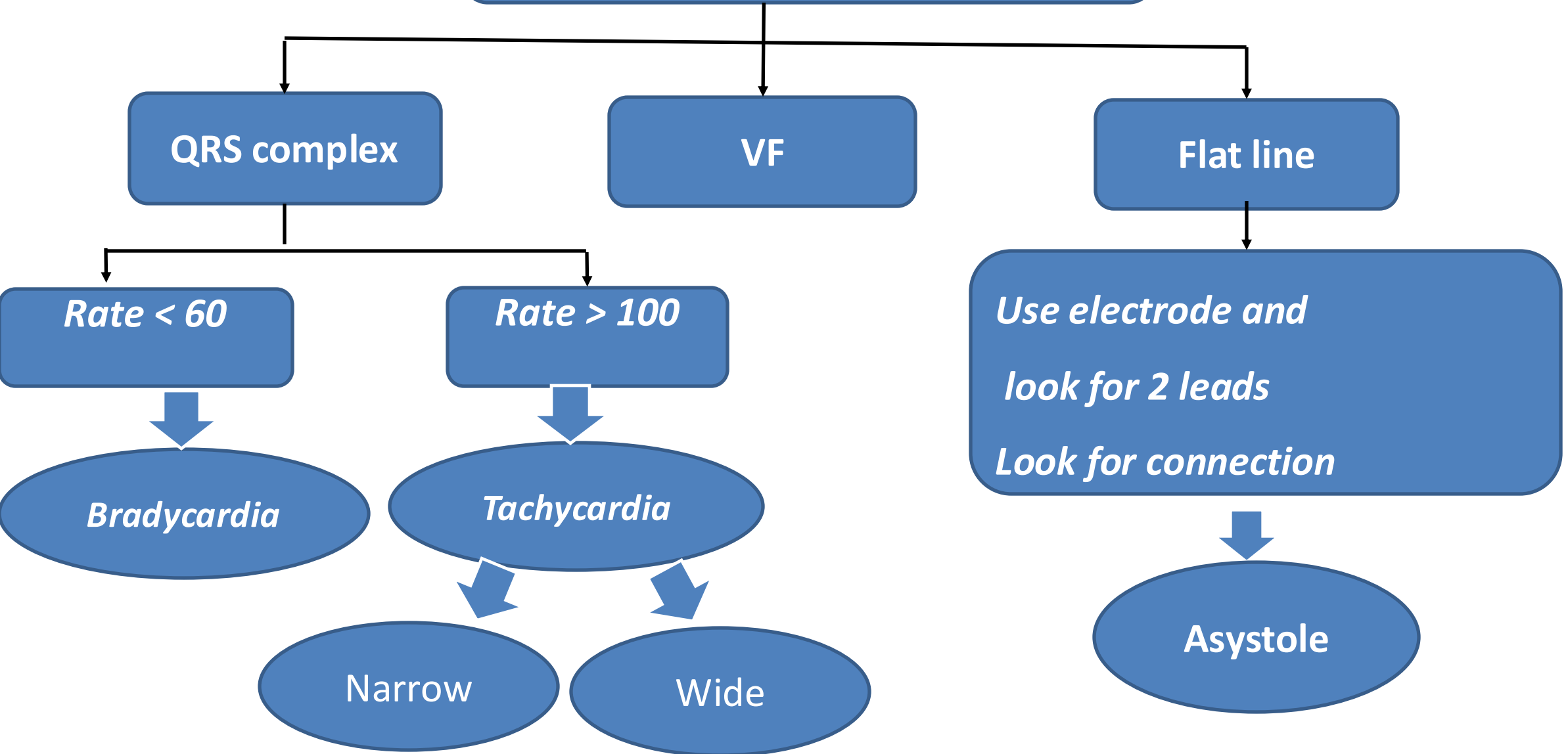


Normal Conduction





Rhythm





Classification of arrhythmia

- **Bradyarrhythmia**
- **Tachyarrhythmia**



มหาวิทยาลัยมหิดล
คณะแพทยศาสตร์
ศิริราชพยาบาล



Bradyarrhythmias

Pattanaporn Supina RN.

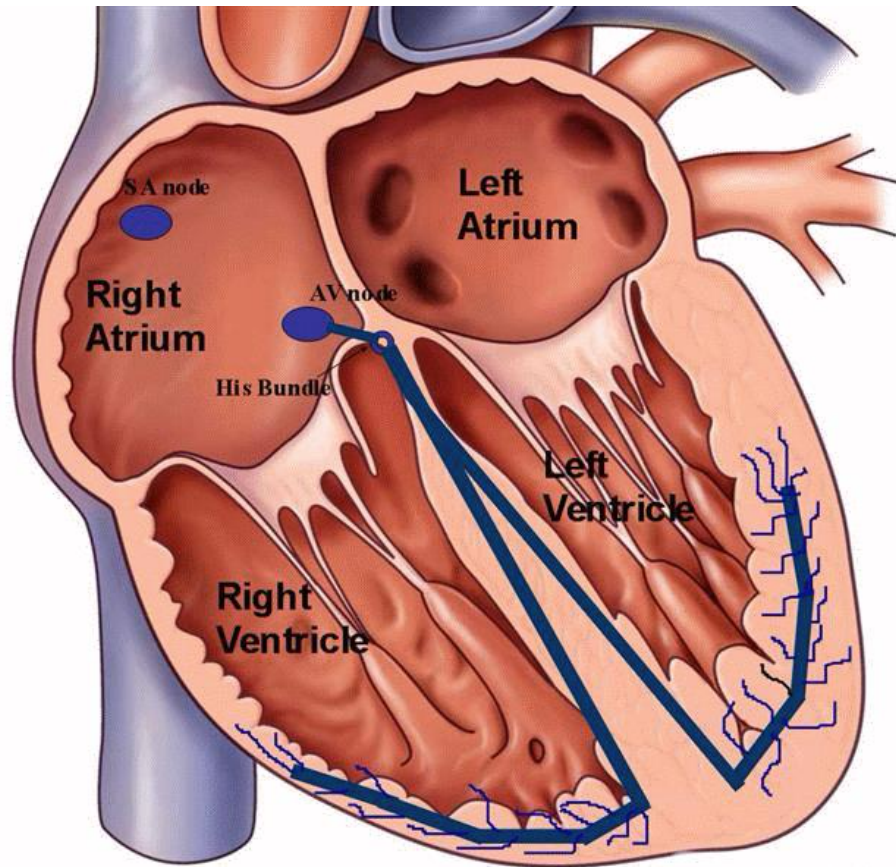
CCU Siriraj Hospital



- **How to interpret Bradyarrhythmias**
- **Nursing care & Management**



Conduction System



SA node
60 -100 BPM

AV node
40 -60 BPM

Ventricular < 40 BPM



Types of Bradyarrhythmias

- **SA node**
 - Sinus pause & Sinus arrest
 - Sinus block
- **AV node**
 - Junctional rhythm
 - AV block
- **Ventricular**
Idioventricular rhythm



Sinus pause & Sinus arrest



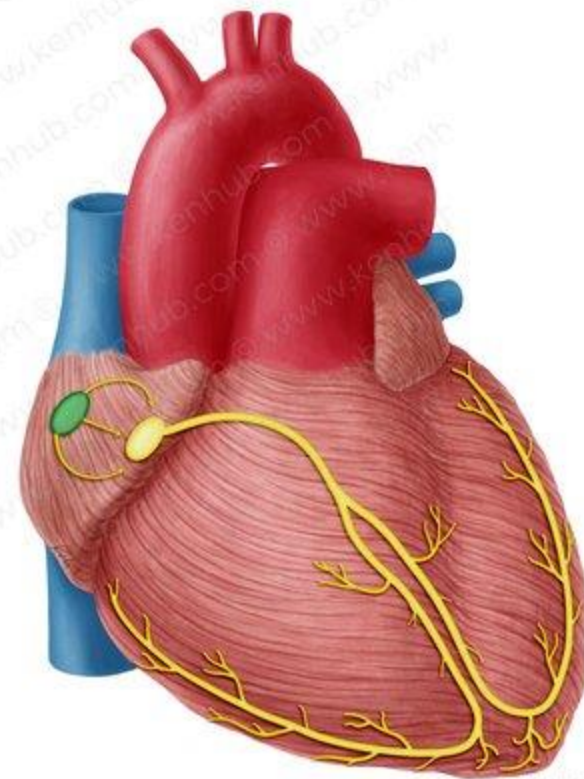
Sinus Pause & Sinus Arrest

SA node หยุดสร้างกระแสไฟฟ้า



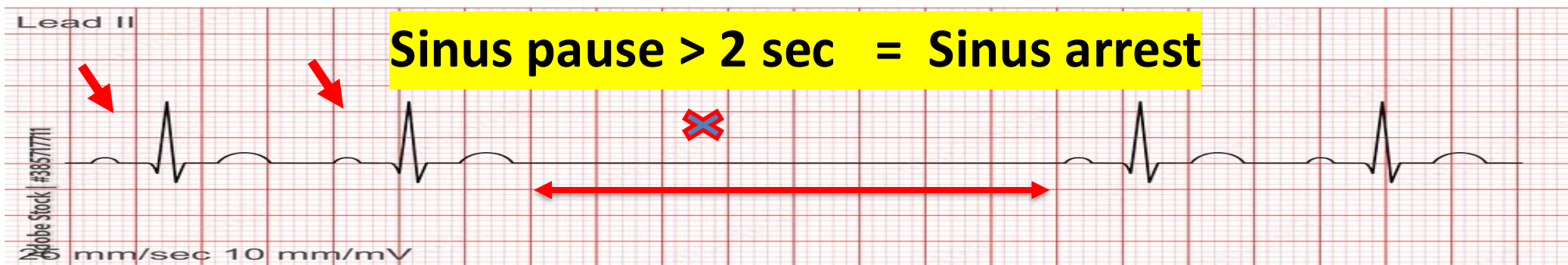
ไม่มี

P & QRS & T wave





Sinus Pause & Sinus Arrest



Rhythm	Rate	P wave	PR interval	QRS
Irregular	Vary but is generally slow or normal	- Absent during pause or arrest - Not multiple of P-P interval	Normal or prolong and constant.	- Usually narrow - Not multiple of R- R interval



Sinus block



Sinus block (SA block)

Lead II



Rhythm	Rate	P wave	PR interval	QRS
Irregular	Vary but is generally slow or normal	- Absent during SA block - Multiple of P-P interval	Normal or prolong and constant.	- Usually narrow - Multiple of R-R interval



Types of Bradyarrhythmias

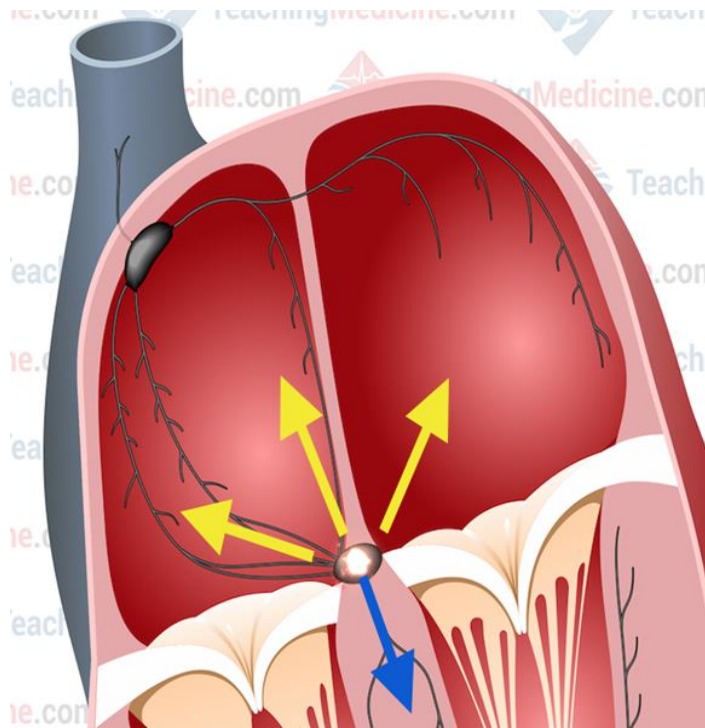
- **SA node**
 - Sinus pause & Sinus arrest
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 - Junctional rhythm
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 - Idioventricular rhythm



Junctional Rhythm



Junctional Rhythm

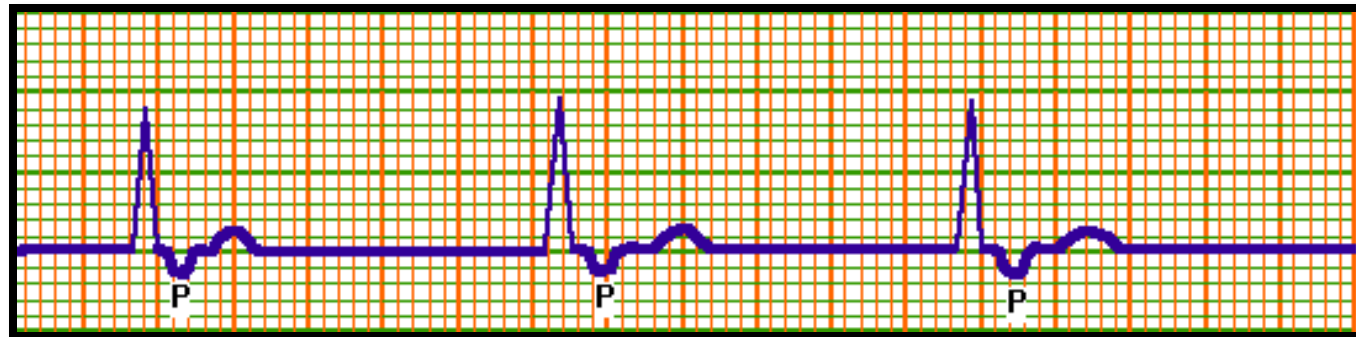
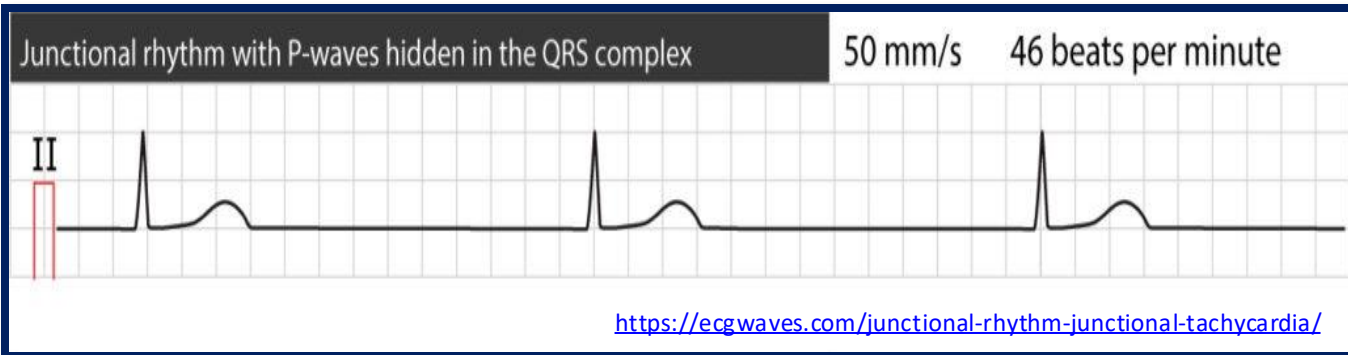
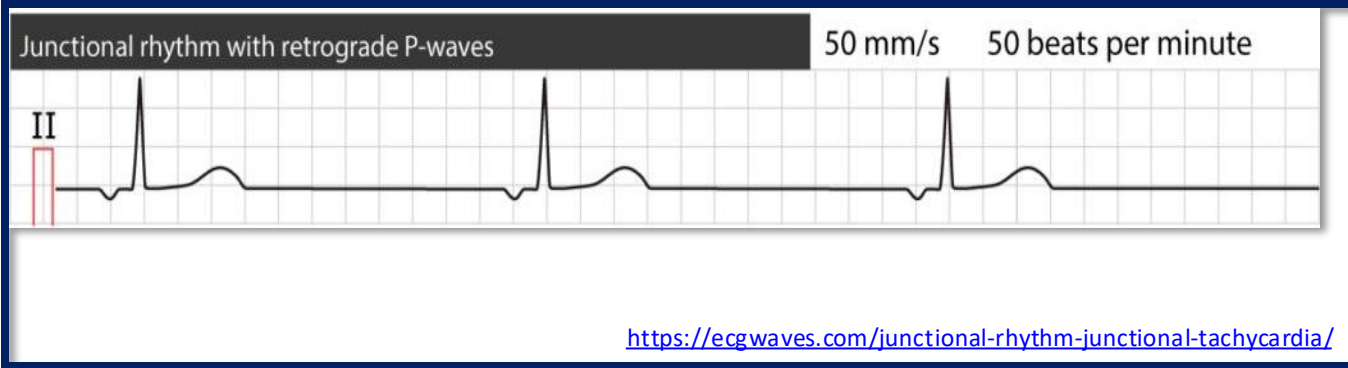
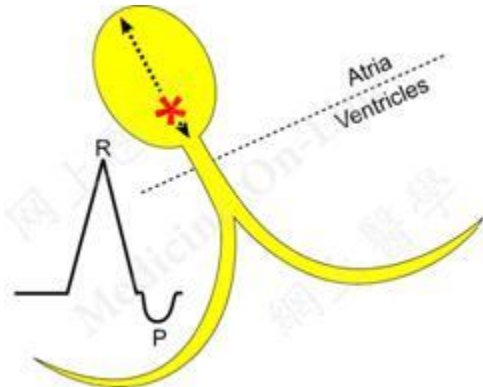
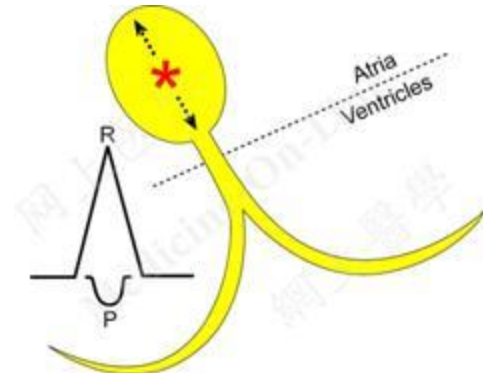
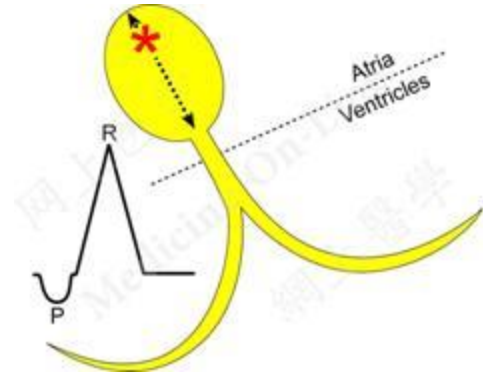


https://www.teachingmedicine.com/Lesson.aspx?l_id=34

Rhythm	regular
Rate	40 - 60 BPM
P wave	หวักลับใน lead II , ไม่มีหรือตามหลัง QRS
PR interval	< 0.12 sec หรือวัดไม่ได้
QRS	< 0.12 sec



Junctional Rhythm





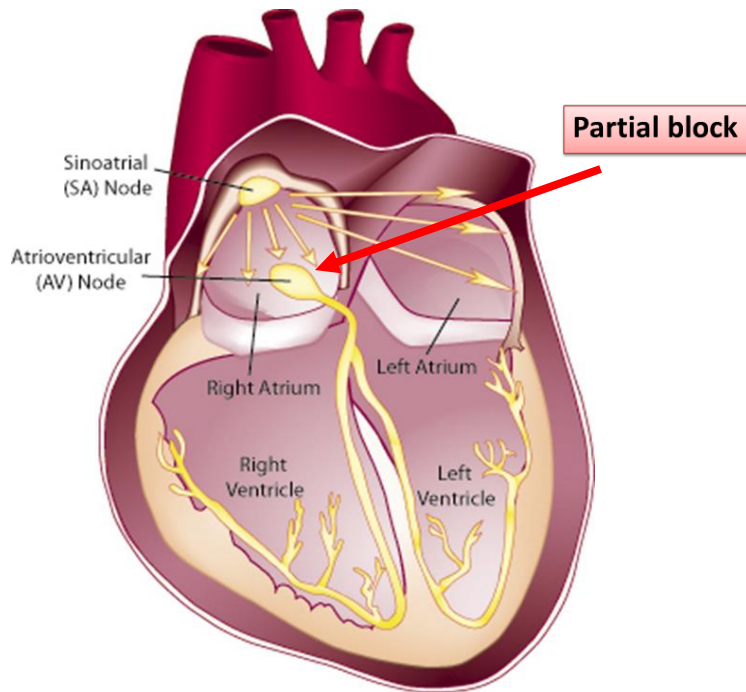
Atrioventricular Block (AV Block)



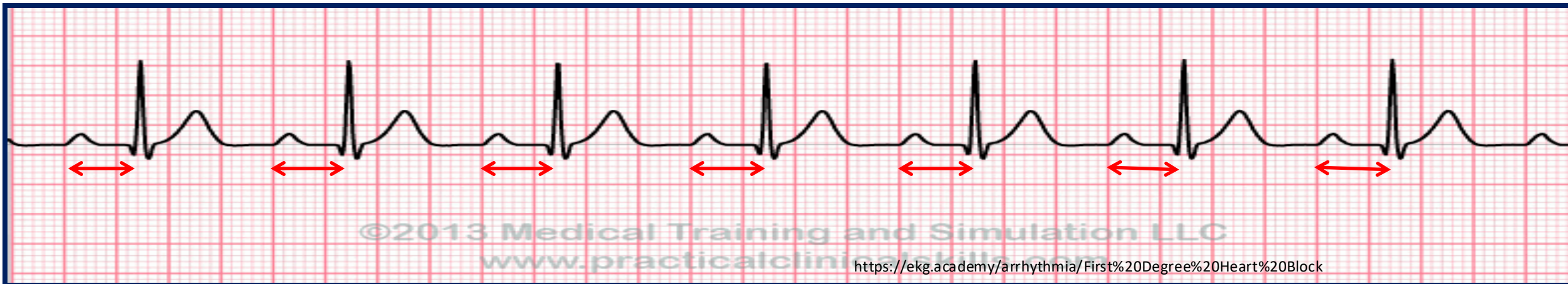
- **First degree AV block**
- **Second degree AV block Mobitz type I (Wenckebach)**
- **Second degree AV block Mobitz type II**
- **Third degree AV block (Complete AV block)**



First degree AV block



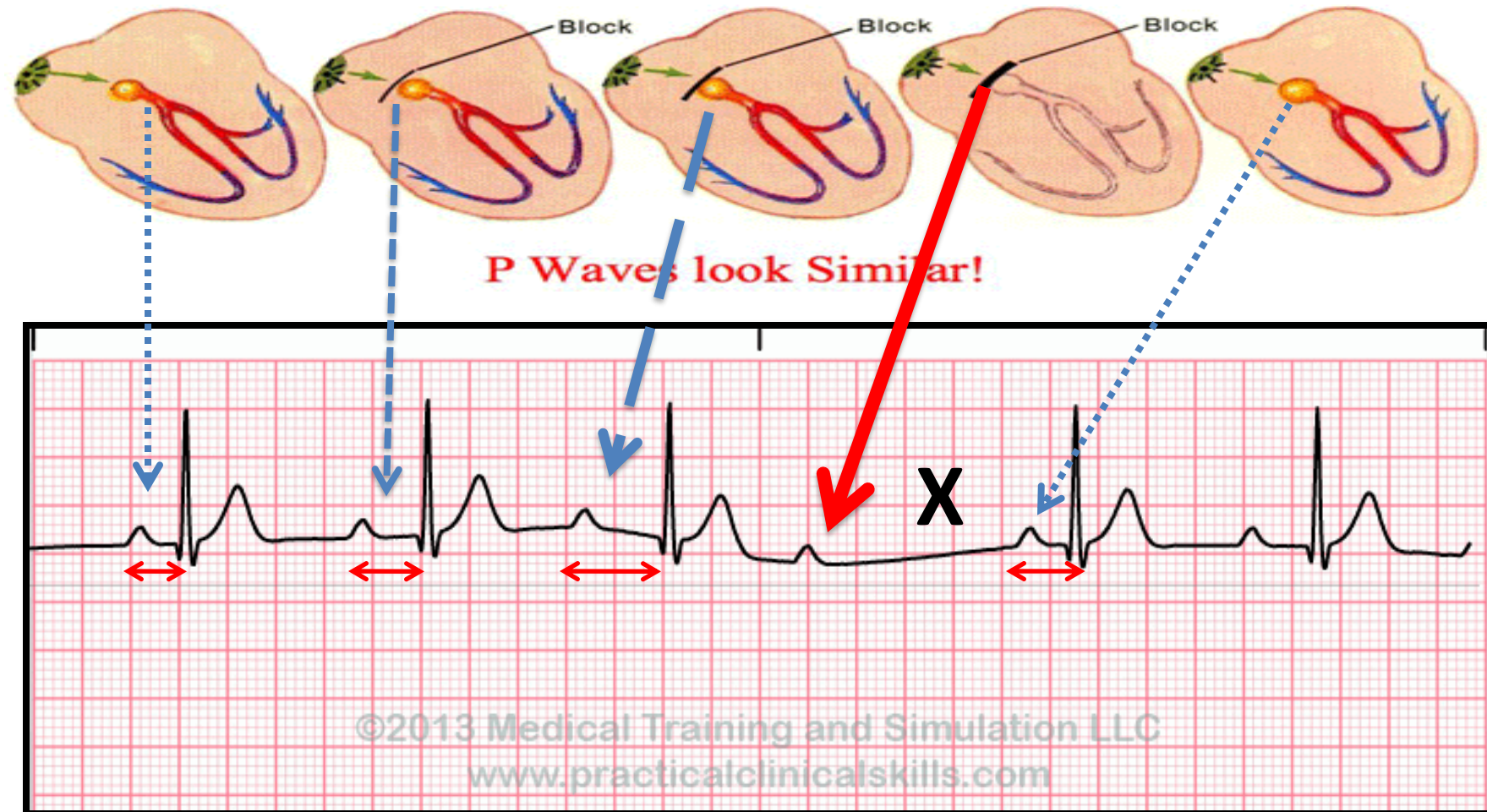
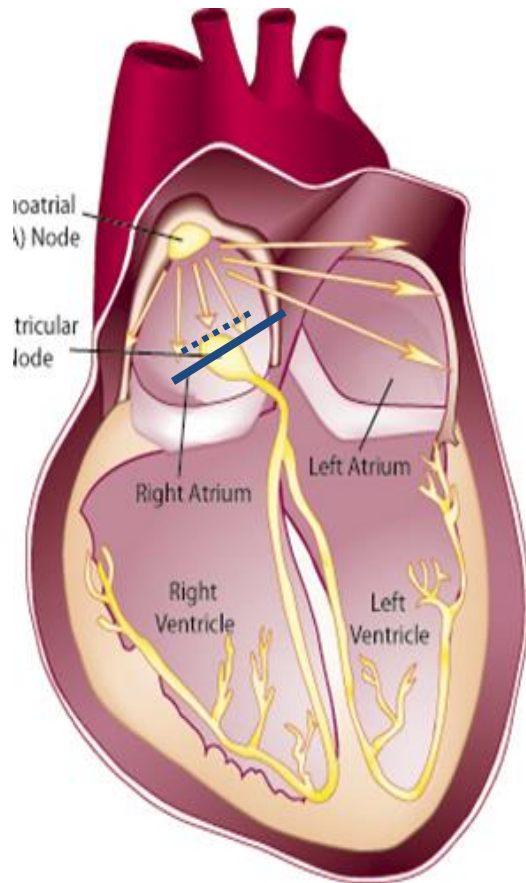
Rhythm	regular
Rate	ขึ้นอยู่กับ rhythm เดิม
P wave	หัวตั้งใน lead II
PR interval	>0.20 sec และคงที่
QRS	<0.12sec ตามหลัง P wave ทุกตัว





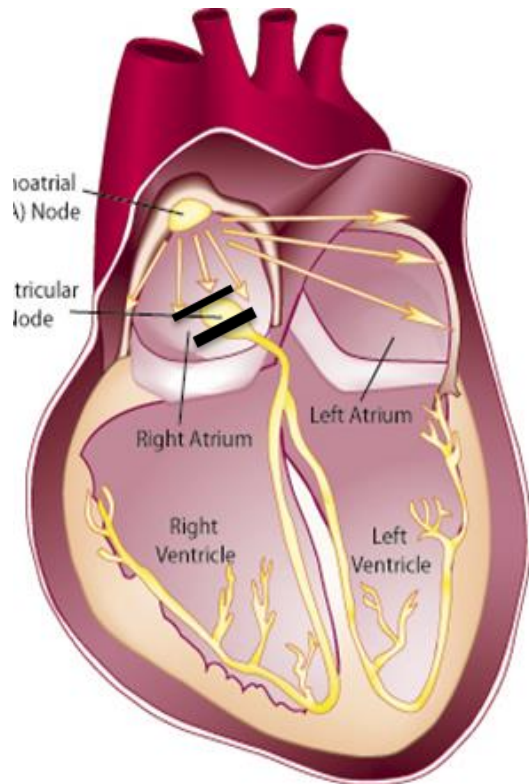
2° AV block Mobitz I (Wenckebach)

2° AV Block Mobitz I





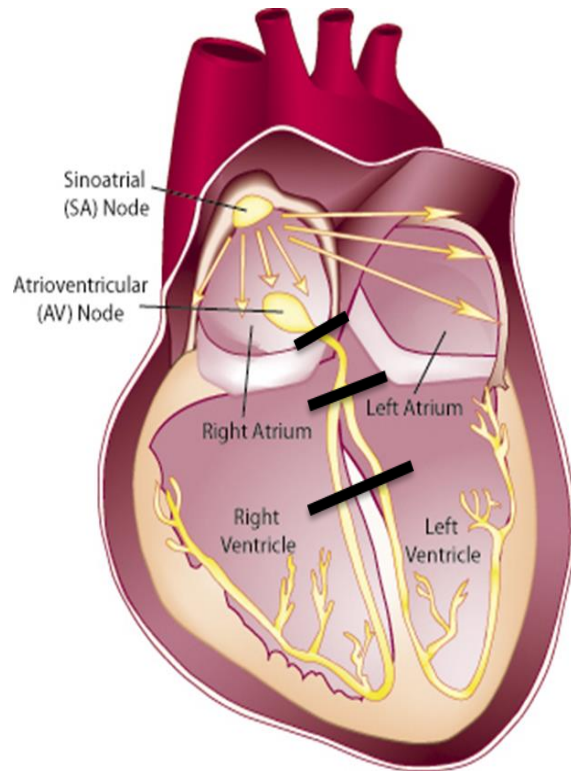
2^oAV block Mobitz I (Wenckebach)



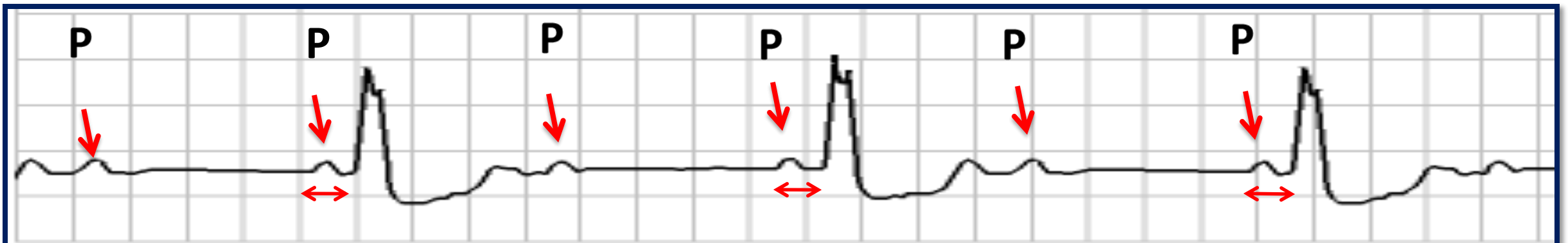
Rhythm	irregular
Rate	atrial rate ปกติ แต่ ventricular rate จะช้ากว่า atrial rate
P wave	หัวตั้งใน lead II
PR interval	ค่อยๆ prolong จนถึง P wave ไม่มี QRS ตาม
QRS	<0.12sec
Notes	PR interval หลัง block จะสั้นกว่า PR interval ก่อนblock



2^oAV block Mobitz II

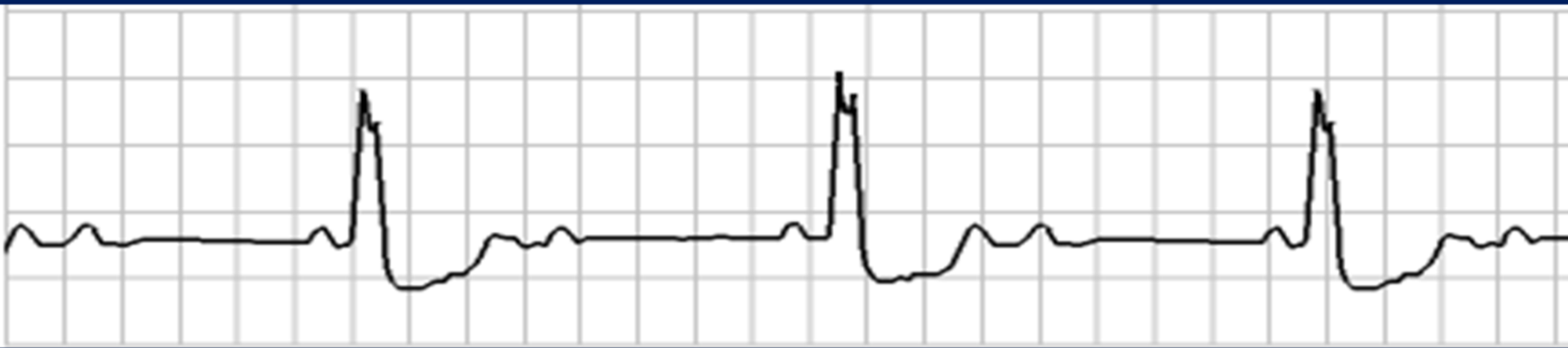
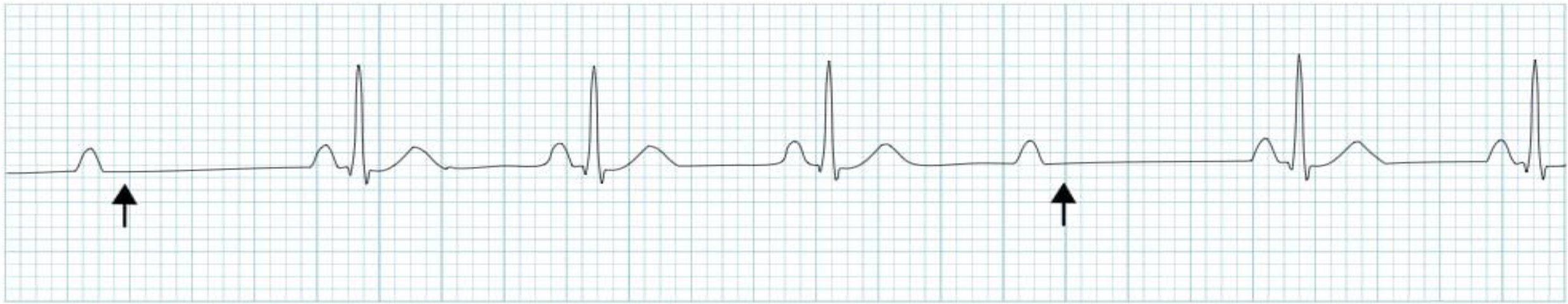


Rhythm	Regular/irregular ขึ้นกับอัตราส่วนการ block ว่าสม่ำเสมอหรือไม่
Rate	atrial rate ปกติ แต่ ventricular rate จะช้ากว่า atrial rate
P wave	หัวตั้งใน lead II
PR interval	มีค่าคงที่
QRS	<0.12sec หรือ > 0.12 sec.
Notes	ต่างกับ Mobitz I คือ จะไม่มีการค่อย ๆ ยาวของ PR interval



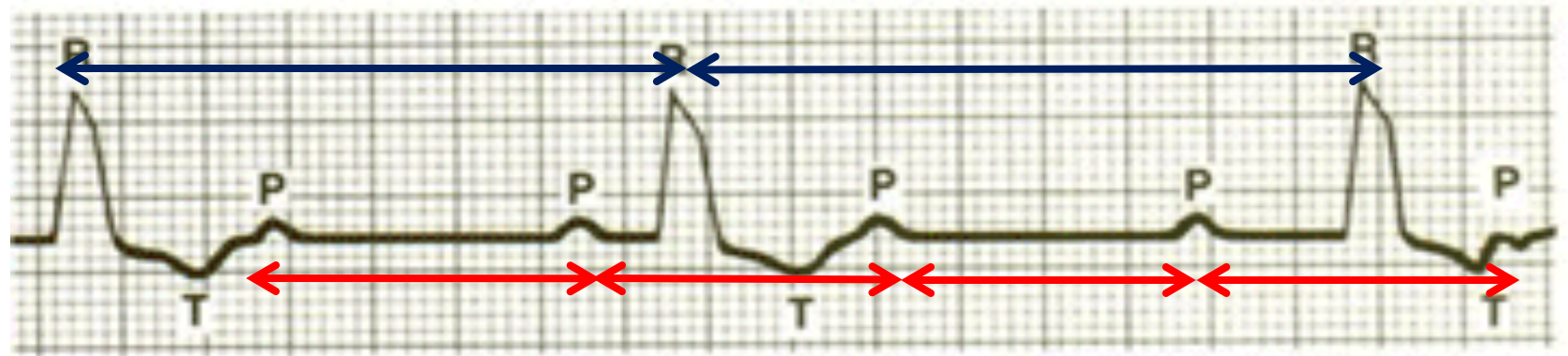
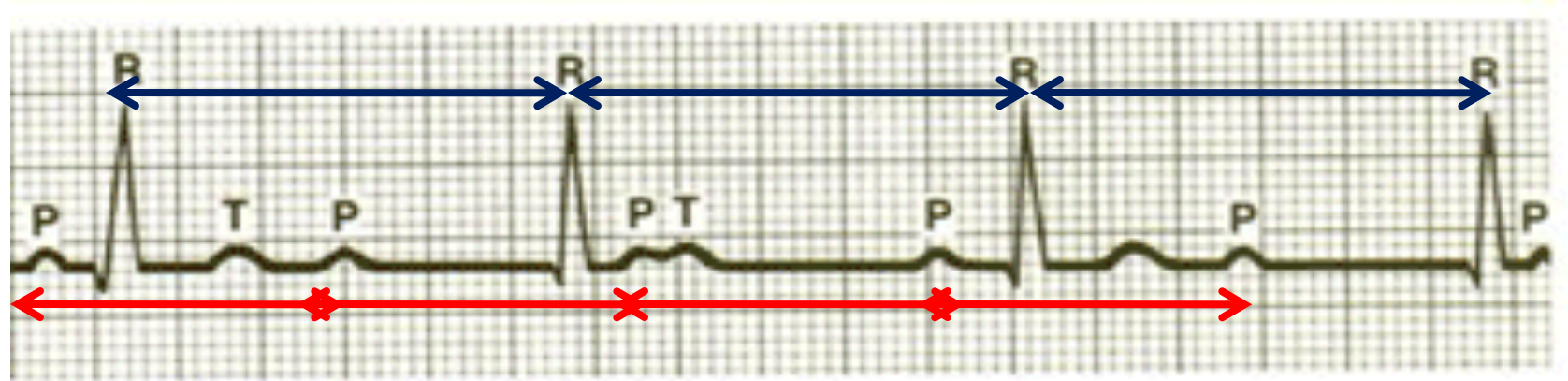
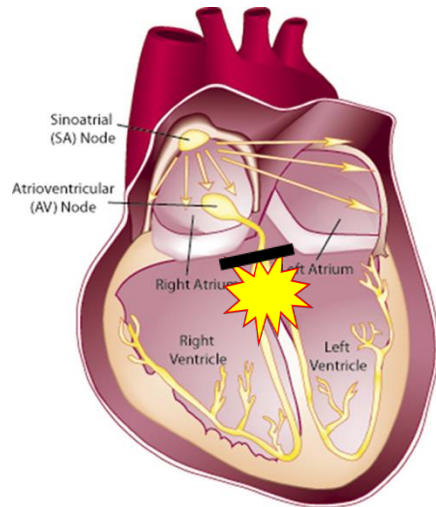
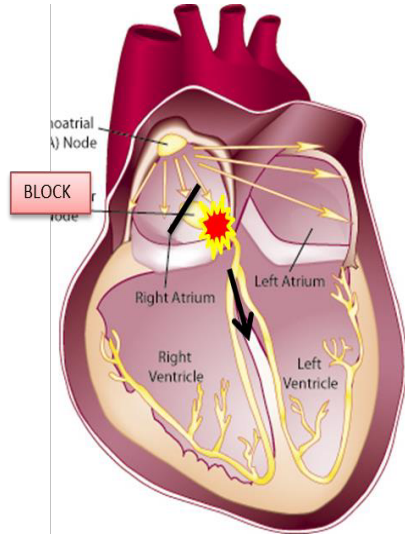


2^oAV block Mobitz II



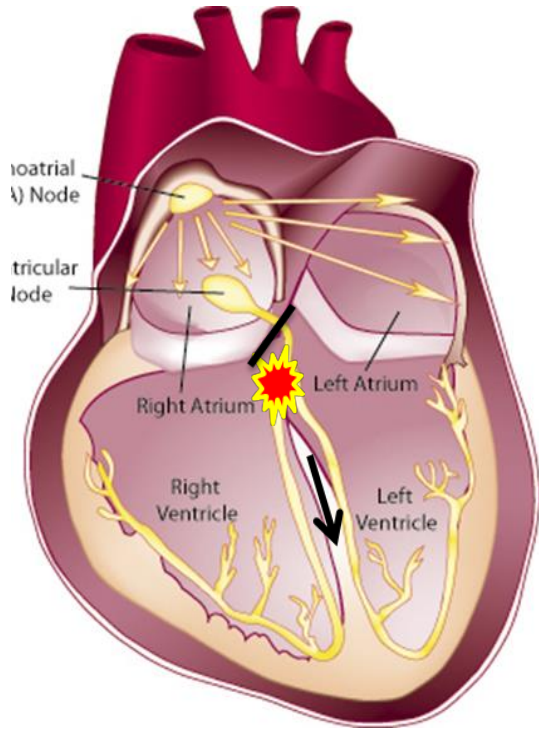


Third degree AV block (complete AV block)

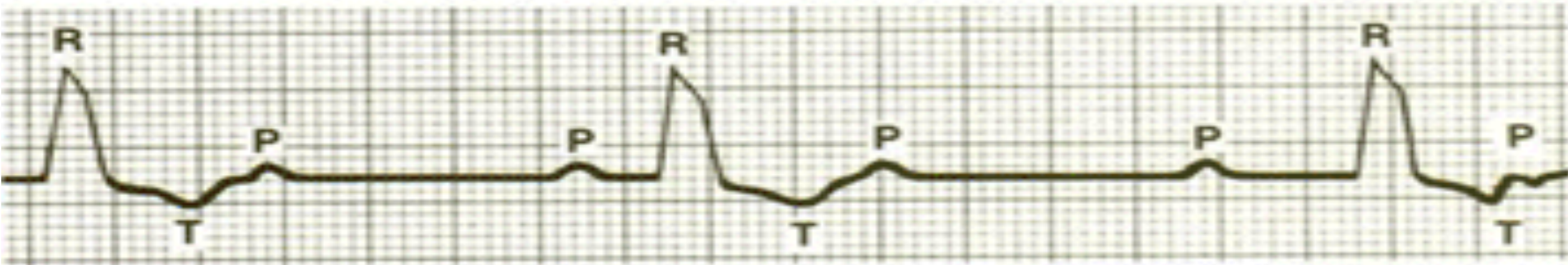




3⁰degree AV block (Complete AV block)



Rhythm	Regular (ทั้ง P-P interval และ R-R interval)
Rate	atrial rate ปกติ แต่ ventricular rate ประมาณ 20-60 BPM
P wave	หัวตั้งใน lead II
PR interval	ไม่มี
QRS	<0.12sec ถ้าเกิดจาก junctional pacemaker >0.12sec ถ้าเกิดจาก ventricular pacemaker
Note	Atria และ ventricle ต่างก็มี pacemaker คนละที่และทำงานไม่เกี่ยวข้องกัน



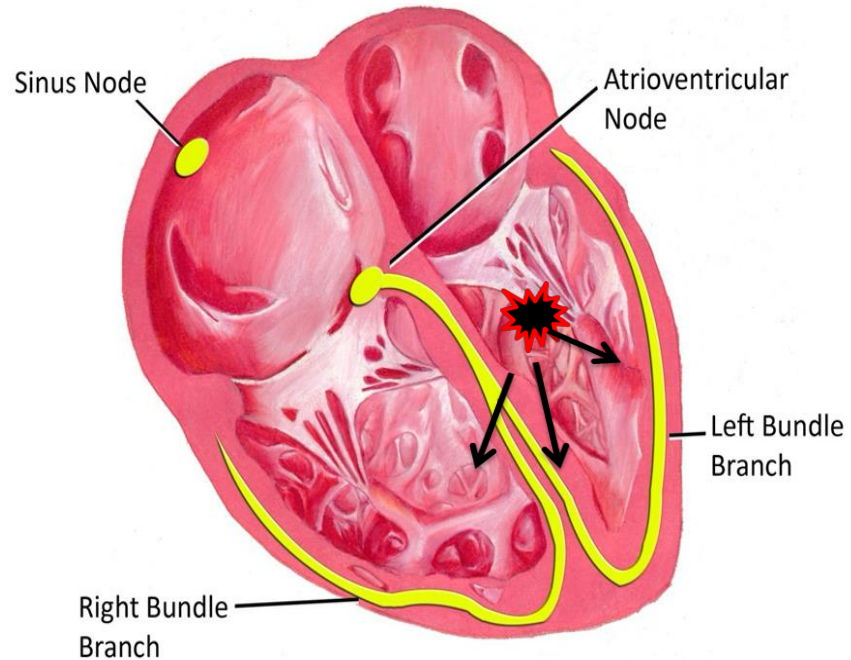


Types of Bradyarrhythmias

- **SA node**
 - Sinus pause & Sinus arrest
 - Sinus block
- **AV node**
 - Junctional rhythm
 - AV block
- **Ventricular**
Idioventricular rhythm

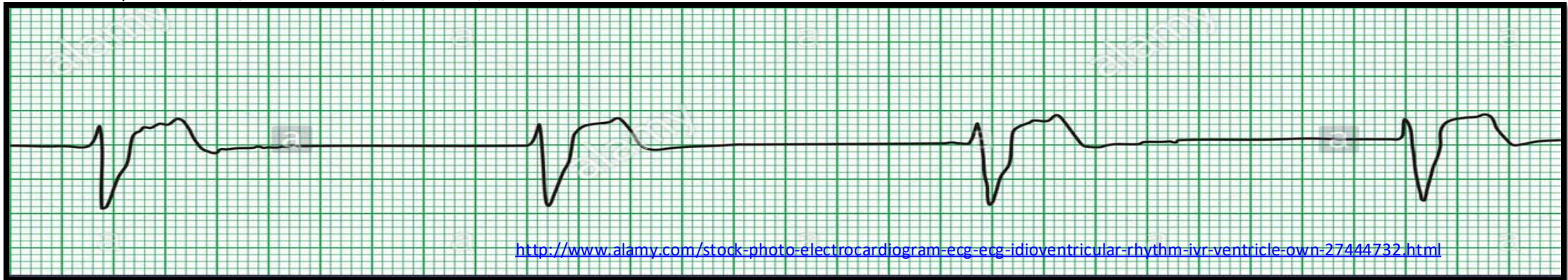


Idioventricular rhythm



Rhythm	regular
Rate	20 – 40 BPM. หรือช้ากว่านี้ได้
P wave	ไม่มี
PR interval	ไม่มี
QRS	> 0.12 sec

<https://pediatricheartspecialists.com/heart-education/14-normal/153-normal-electrical-conduction>



<http://www.alamy.com/stock-photo-electrocardiogram-ecg-ecg-idioventricular-rhythm-ivr-ventricle-own-27444732.html>

Common causes of bradyarrhythmia

- Drugs; digitalis, beta blockers, amiodarone
- Infarction/Ischemia; Inferior wall MI
- Electrolytes abnormalities; Hyperkalemia
- Infection of heart tissue (myocarditis)
- Hypothermia
- A complication of heart surgery
- Hypothyroidism
- Heart tissue damage related to aging

D
I
E



- **Assessment**
- **Intervention**
- **Evaluation**



- **Assessment**

- History; Underlying disease
- Physical ; V/S, Symptomatic bradycardia ??
 - ❖ Hypotension
 - ❖ Acute altered mental status
 - ❖ Ongoing chest pain
 - ❖ HF
 - ❖ Signs of shock.



- **Assessment**
- **Intervention**
- **Evaluation**



- **Intervention**

- Oxygen support
- Monitor EKG, EKG 12 leads
- IV Access → LAB; blood for electrolyte, Magnesium, Calcium, Cardiac enzymes,
- Clinical stable → monitor and observe
- Withholding or reducing the dose of negative chronotropes ; Digoxin, Beta blocker, Calcium channel Blocker
- Correct LAB or another cause

- If Unstable or Symptomatic bradycardia ????



- ***Medication***

- ***Atropine 1 mg. IV q 3 -5 min (max 3 mg.)***

- ***Dopamine IV drip***

- ***Adrenaline IV drip***



- Phlebitis

- Extravasation → tissue necrosis



If Unstable or Symptomatic bradycardia ????

Treatment	Nursing Actions	Rationale
<u>Medication</u>		
Atropine	- If given at very low doses (i.e. <0.5 mg) → Bradycardia - Use cautiously in ACS	↑ HR → ↑ Ischemia
- Dopamine - Adrenaline	- Use cautiously in ACS, - Beware Phlebitis, Extravasation	↑ HR, Contractility → ↑ Ischemia - Vasoconstriction

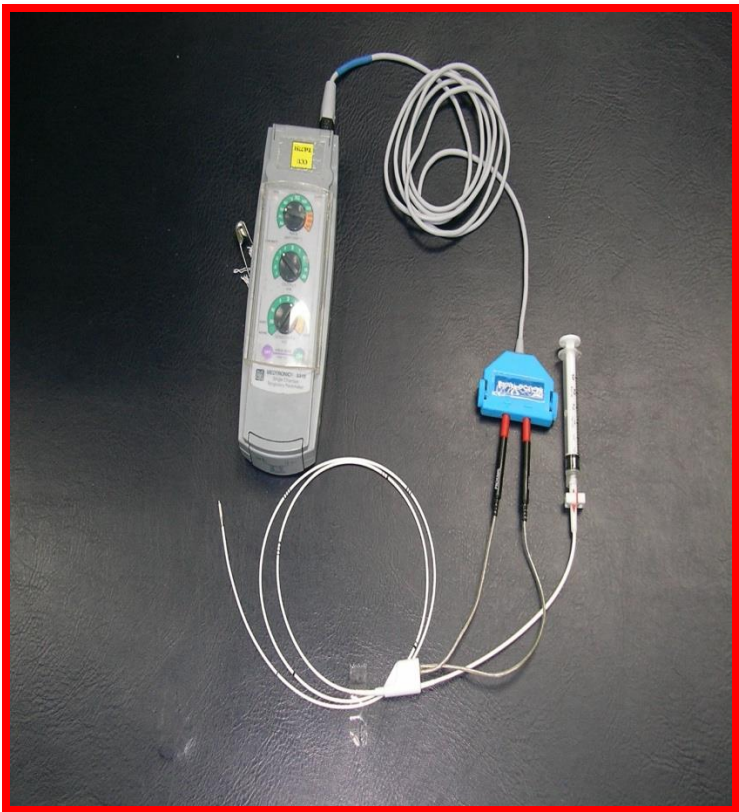


- **Temporary Pacemaker**
 - Transcutaneous temporary pacemaker





- **Temporary Pacemaker**
 - Transvenous temporary pacemaker





- **Intervention**

Treatment	Nursing Actions	Rationale
<u>Temporary Pacemaker</u>	• Inform the patient → Reason, may be discomfort • Analgesia or sedation • Nurses' knowledge and skill regarding temporary pacemaker	

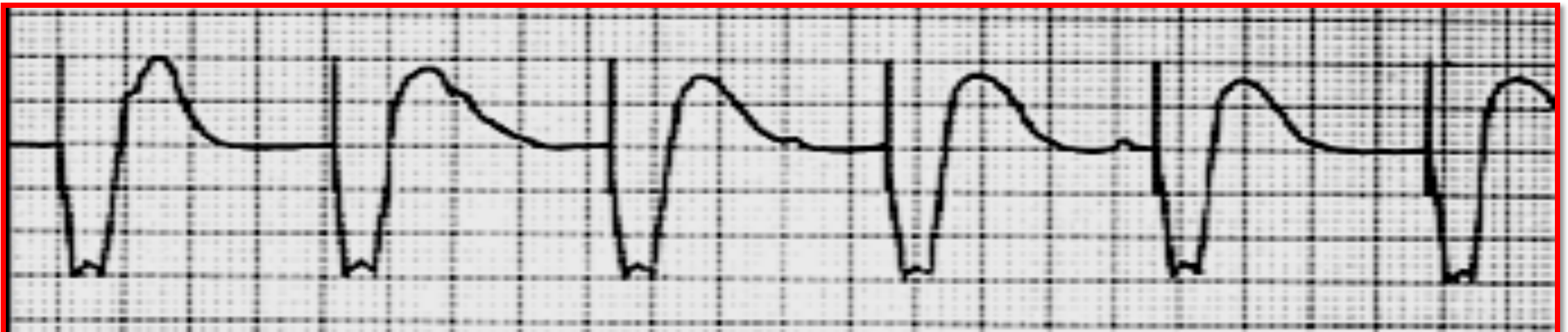


- **Evaluation**

- Vital signs

- Symptoms

Response ???



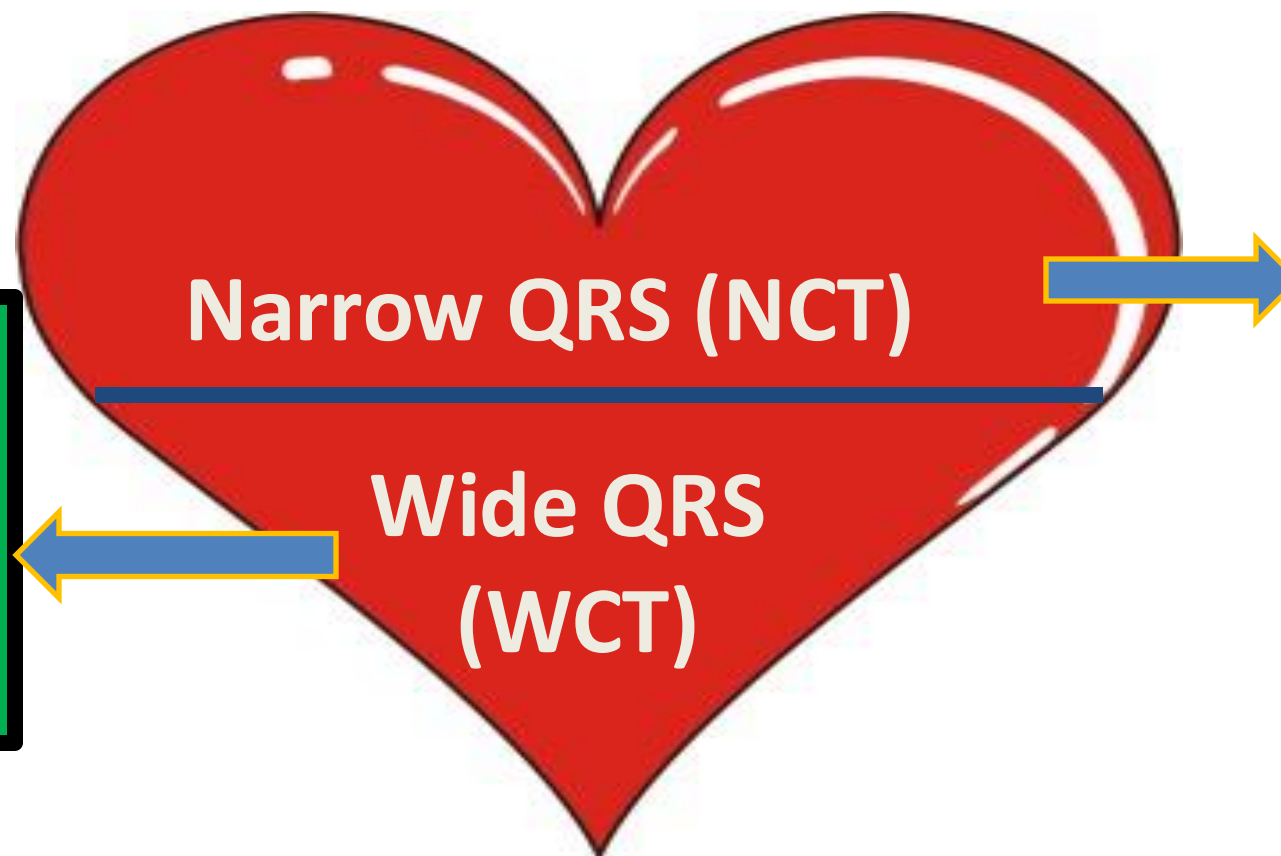


Classification of arrhythmia

- **Bradyarrhythmia**
- **Tachyarrhythmia**



Tachyarrhythmia



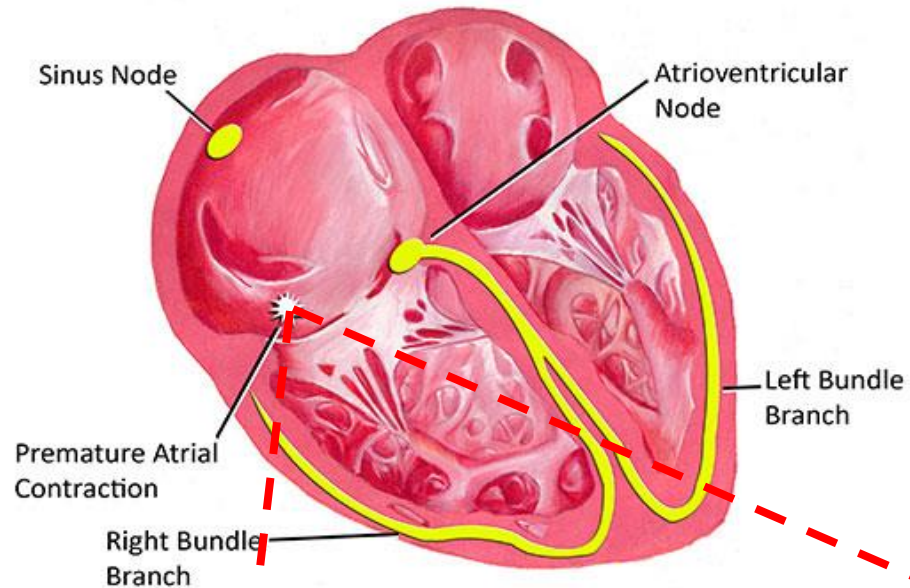
- PVC
- Ventricular Tachycardia (VT)
- Polymorphic VT
- Torsade de pointes
- Ventricular Fibrillation (VF)

- PAC
- SVT
- A. Flutter
- AF
- A. tachycardia
- MAT
- Junctional tachycardia



Premature Atrial Contraction (PAC)

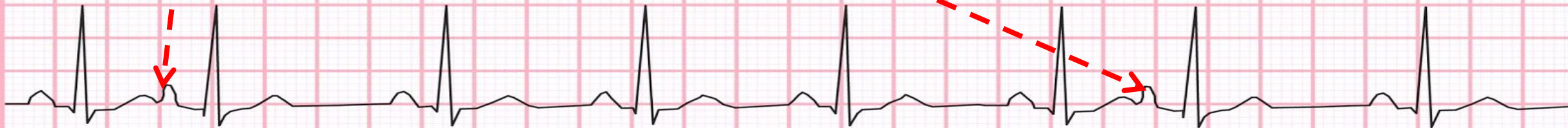
Premature Atrial Contraction



<https://pediatricheartspecialists.com/heart-education/18-arrhythmia/185-premature-atrial-contractions>

Rhythm	ช่วงที่เป็น PAC จะ irregular
Rate	ขึ้นอยู่กับ rhythm เดิม
P wave	ต่างจาก sinus P wave หรืออาจซ้อนซ่อนอยู่ใน T wave
PR interval	0.12 – 0.20 sec
QRS	< 0.12 sec

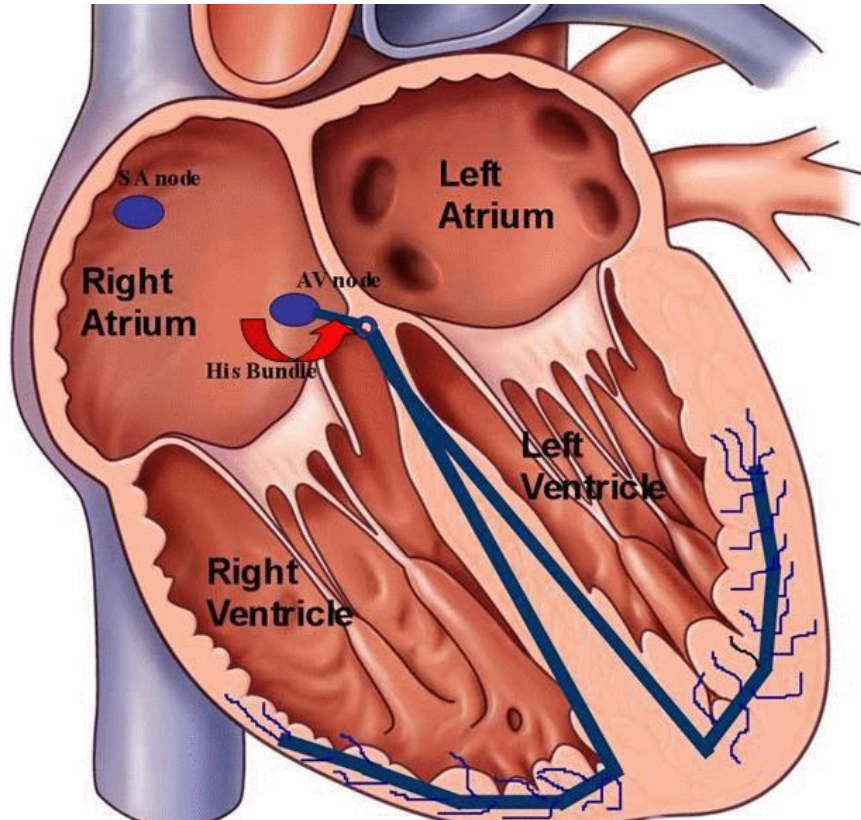
Lead II



<https://ecg-educator.blogspot.com/2016/11/premature-atrial-contraction-pac.html>



Supraventricular tachycardia

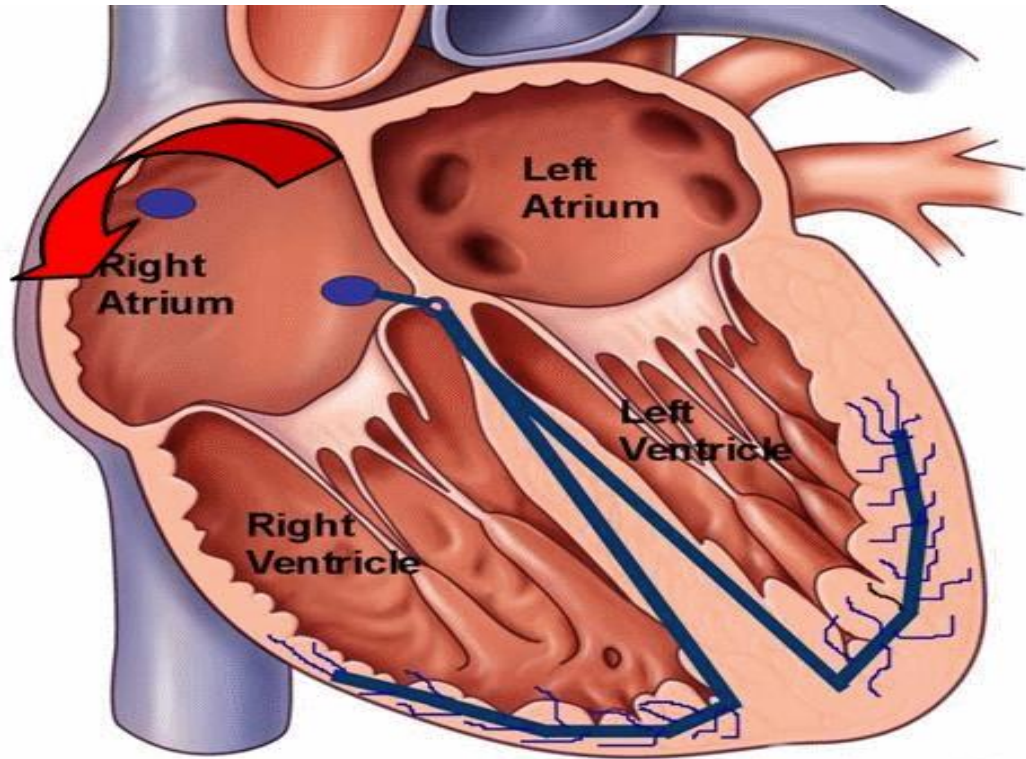


Rhythm	Regular
Rate	150 -250 BPM
P wave	มักมองไม่เห็น P wave
PR interval	ไม่สามารถวัดได้
QRS	<0.12sec

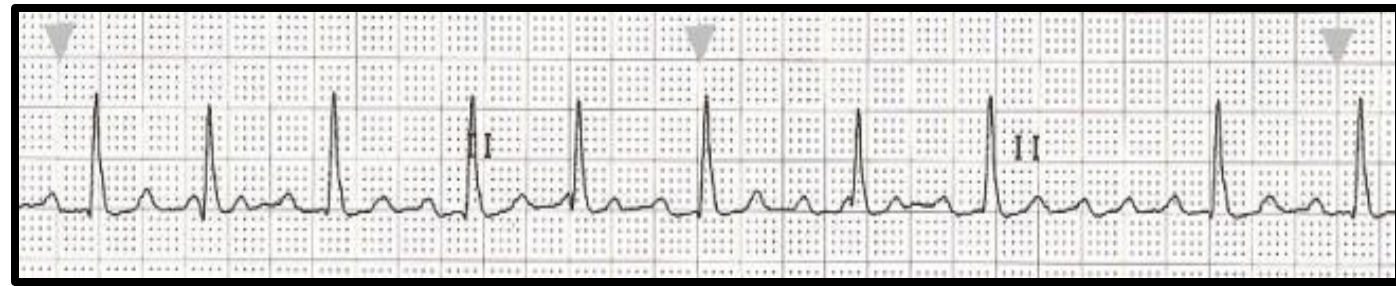




Atrial Flutter

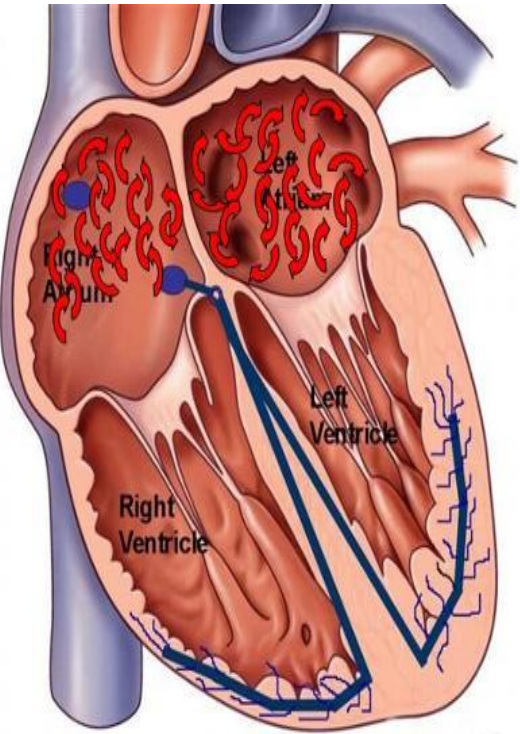


Rhythm	Atrial → Regular Ventricular → Regular or Irregular
Rate	Atrial rate 250 -350 BPM แต่ ventricular rate ไม่แน่นอน
P wave	Sawtooth pattern
PR interval	วัดไม่ได้
QRS	<0.12sec





Atrial Fibrillation

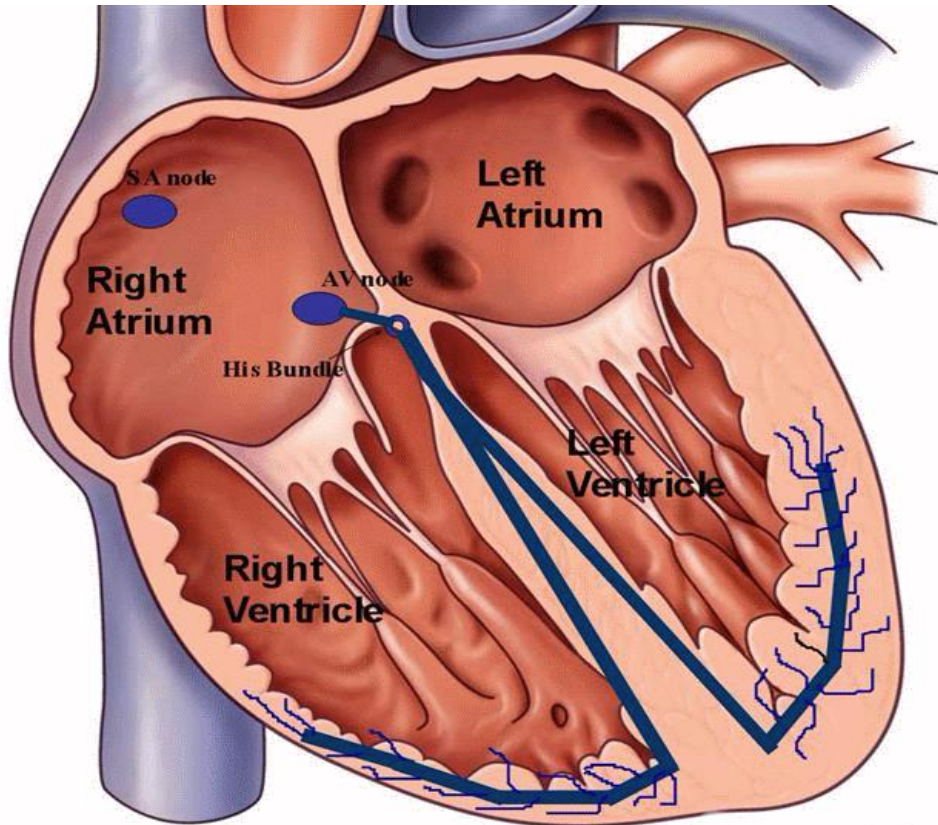


Rhythm	Ventricular → Irregular
Rate	Atrial rate 400 -600 BPM แต่ ventricular rate ไม่แน่นอน
P wave	Fibrillation wave
PR interval	วัดไม่ได้
QRS	<0.12sec

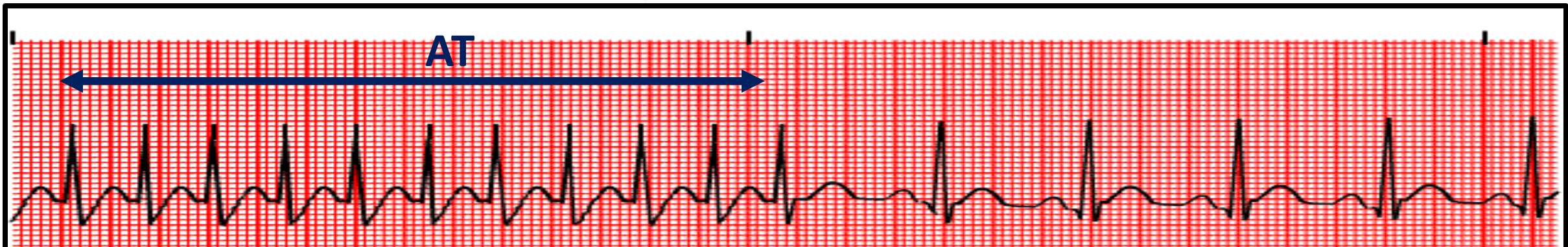




Atrial Tachycardia

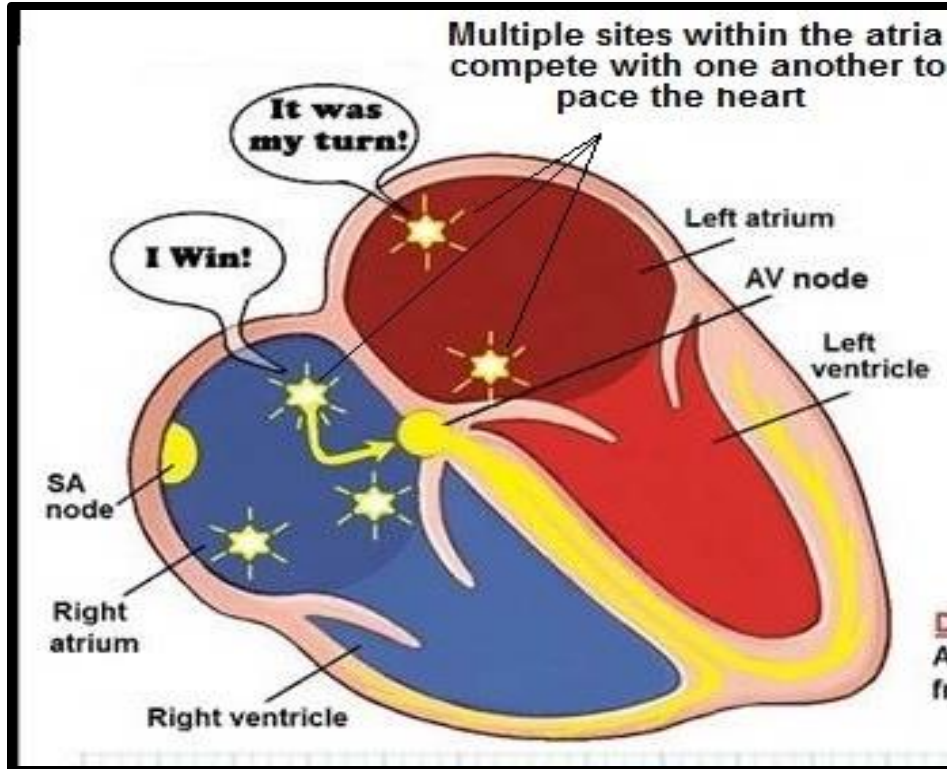


Rhythm	Regular
Rate	150 -250 BPM
P wave	Inverted in lead II,III & aVF
PR interval	0.12 – 0.20sec
QRS	<0.12sec





Multifocal atrial tachycardia

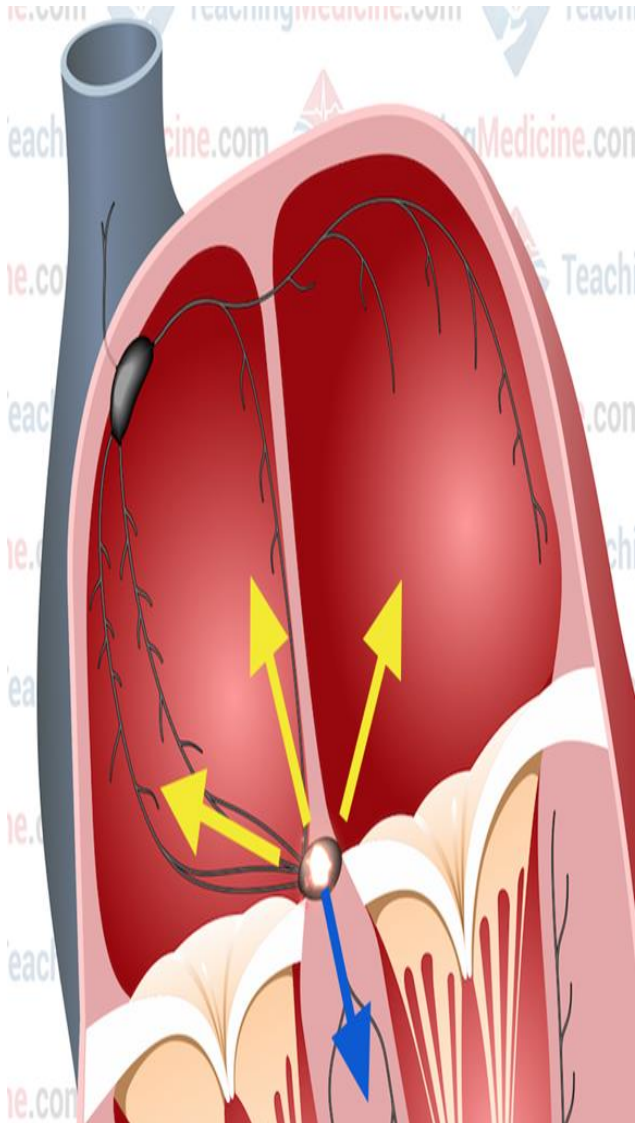


Rhythm	Irregular
Rate	> 100 BPM
P wave	แตกต่างกันอย่างน้อย 3 แบบใน lead เดียวกัน
PR interval	ไม่แน่นอน
QRS	<0.12sec



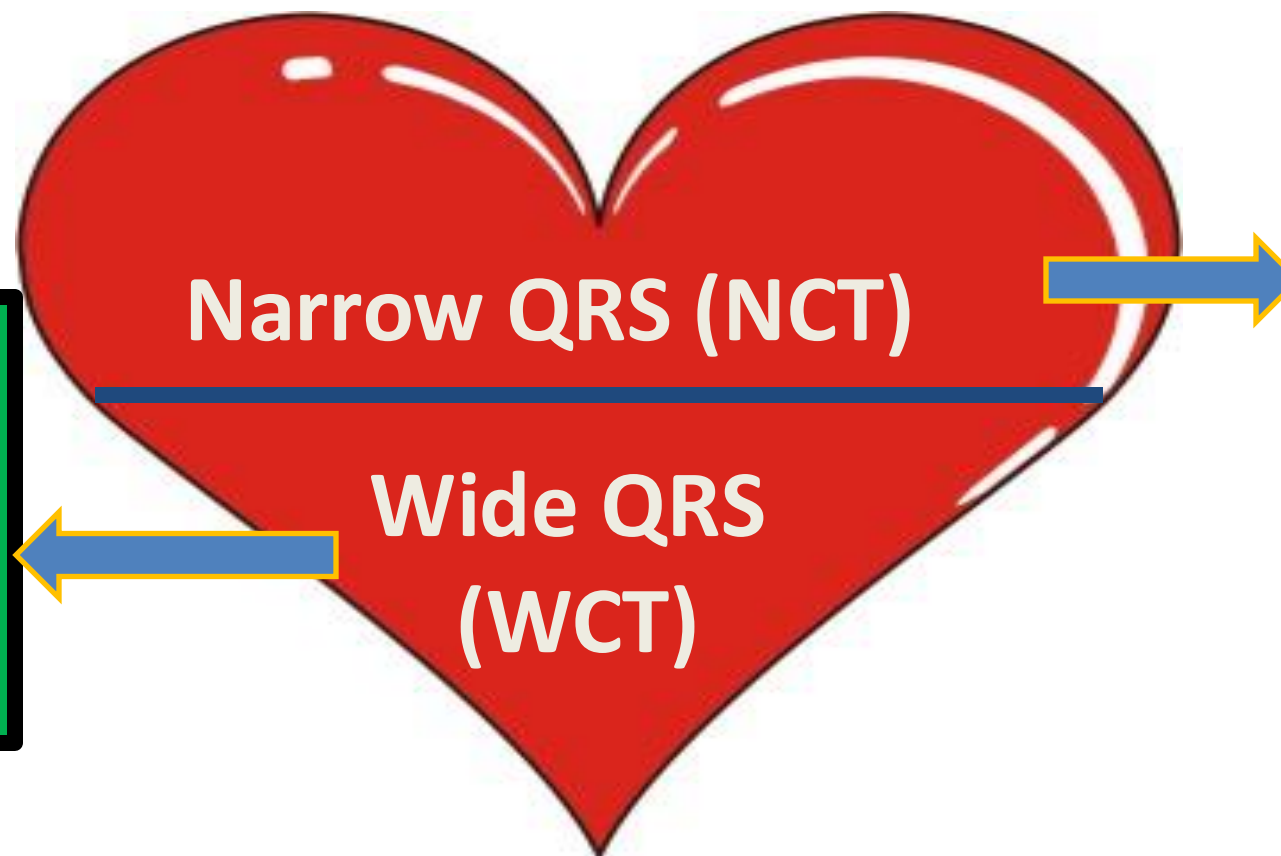


Junctional tachycardia





Tachyarrhythmia



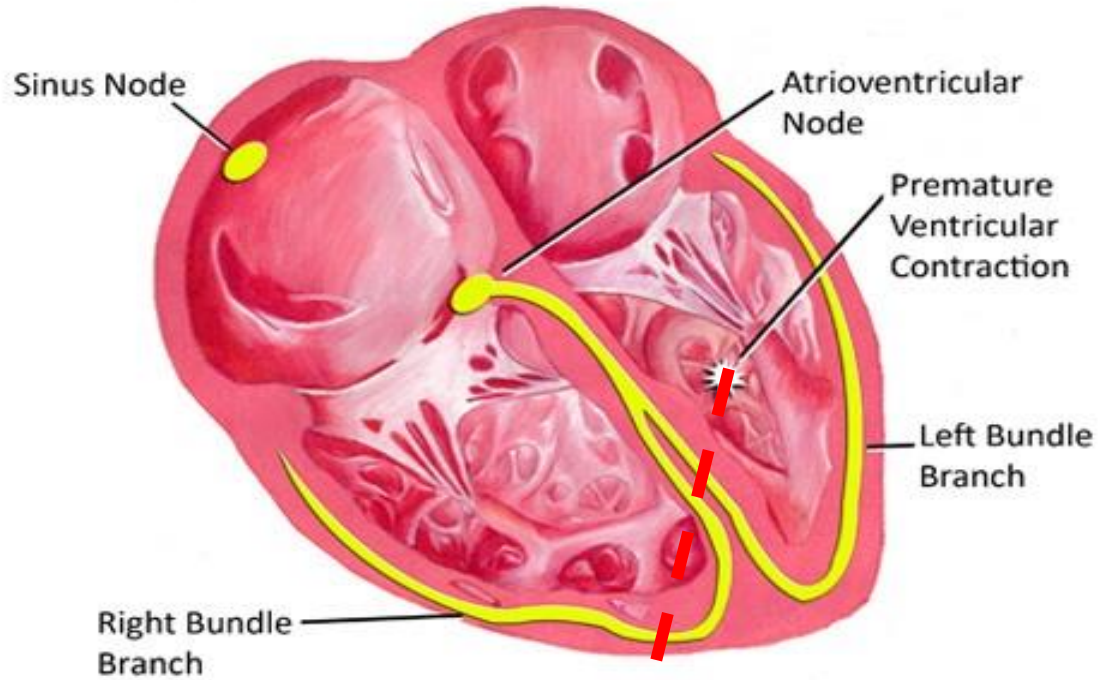
- PAC
- SVT
- A. Flutter
- AF
- A. tachycardia
- MAT
- Junctional tachycardia

- PVC
- Ventricular Tachycardia (VT)
- Polymorphic VT
- Torsade de pointes
- Ventricular Fibrillation (VF)



Premature Ventricular Contraction (PVC)

Premature Ventricular Contraction



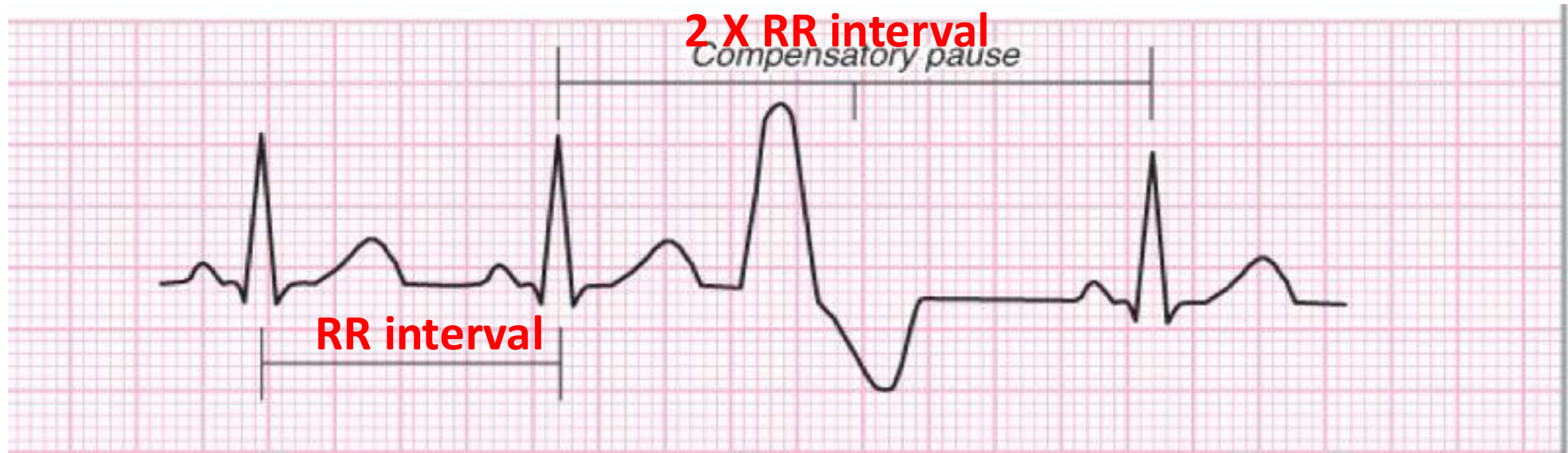
Rhythm	ช่วงที่เป็น PVC จะ irregular
Rate	ขึ้นอยู่กับ rhythm เดิม
P wave	ไม่มี
PR interval	ไม่มี
QRS	> 0.12 sec, รูปร่างต่างจากเดิม
Note	T wave มักมีทิศทางตรงข้าม QRS

Lead II





Compensatory pause





Unifocal PVC

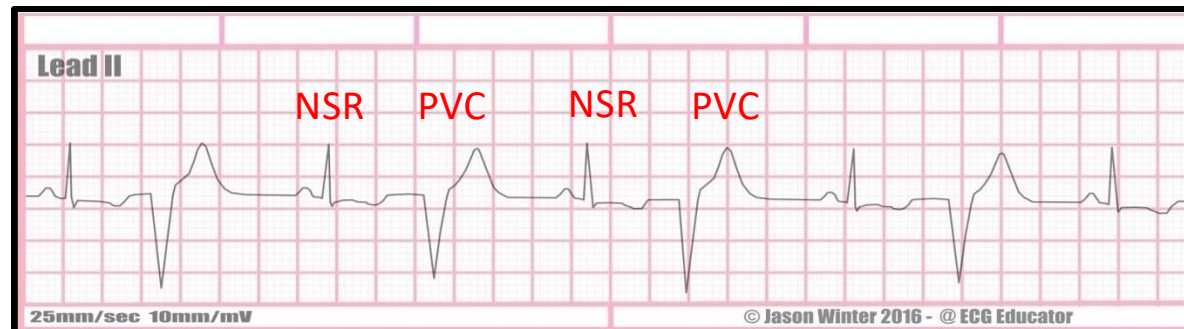


Multifocal PVC

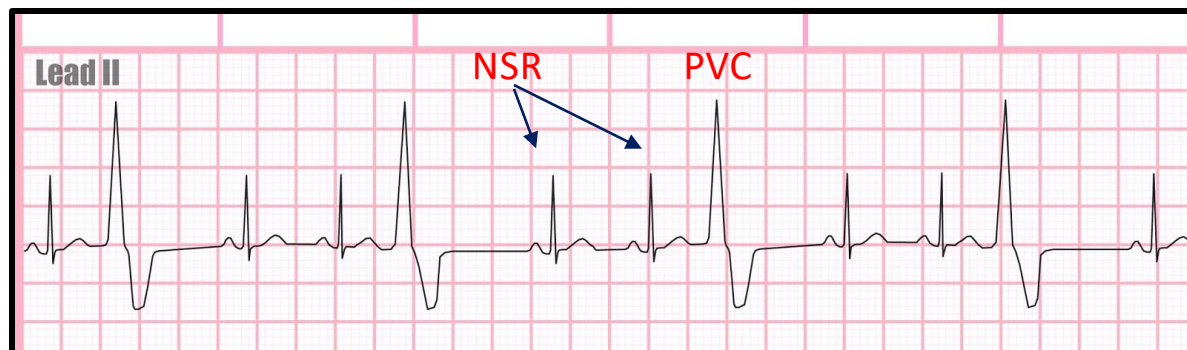




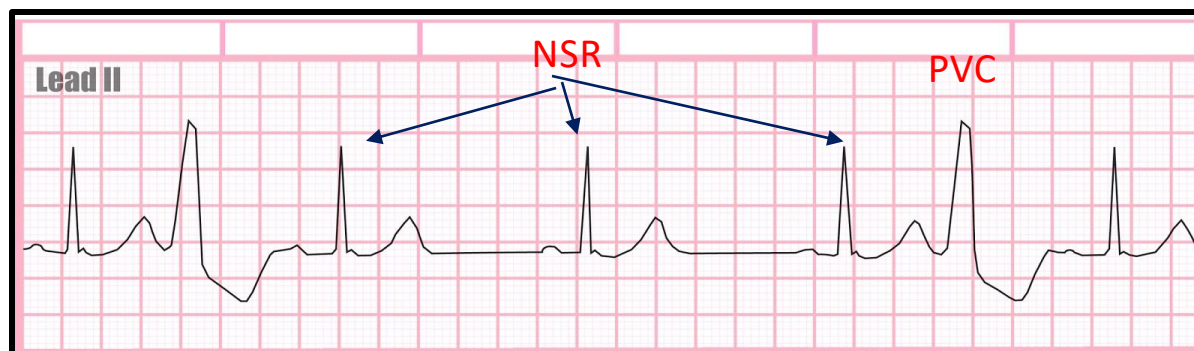
Bigeminy PVC



Trigeminy PVC

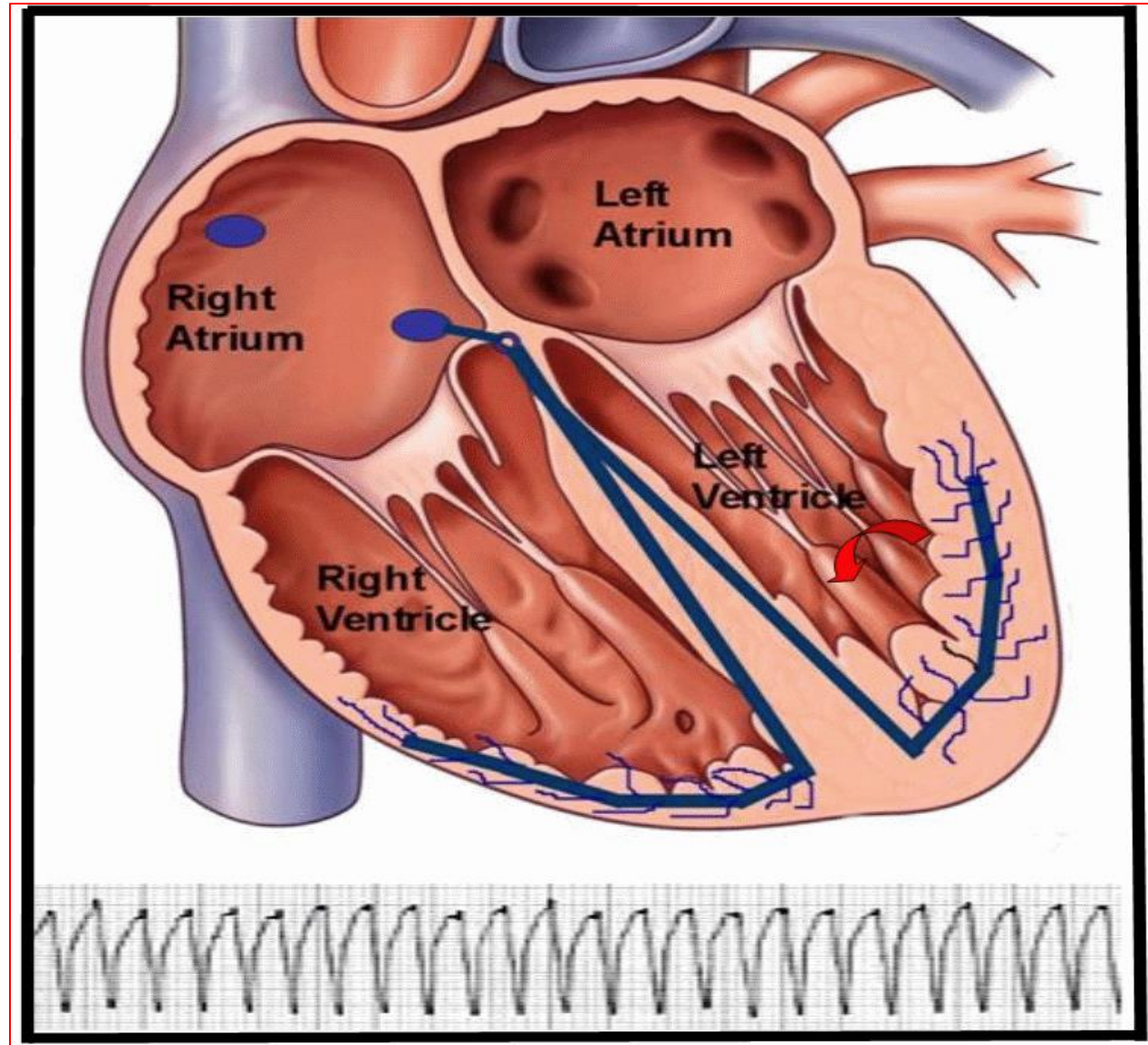


Quadrigeminy PVC





Ventricular tachycardia

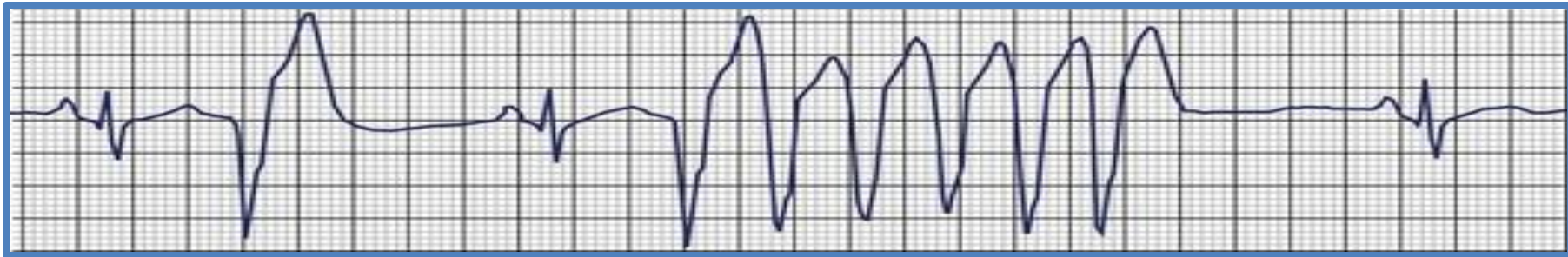




Ventricular tachycardia

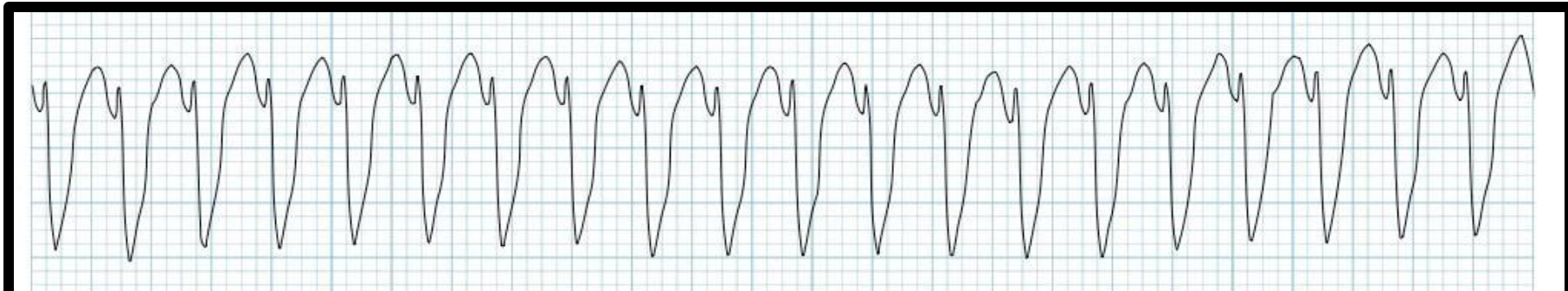
■ Non - Sustained VT

- Episodes last at least 3 beats but < 30 sec



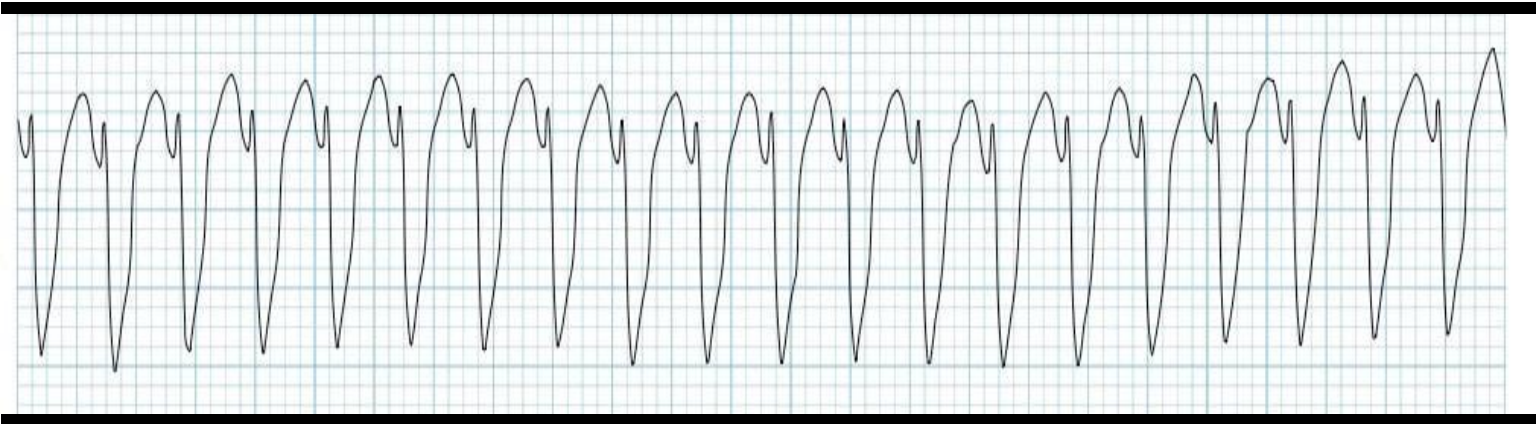
■ Sustained VT

- Episodes last at least 30 sec





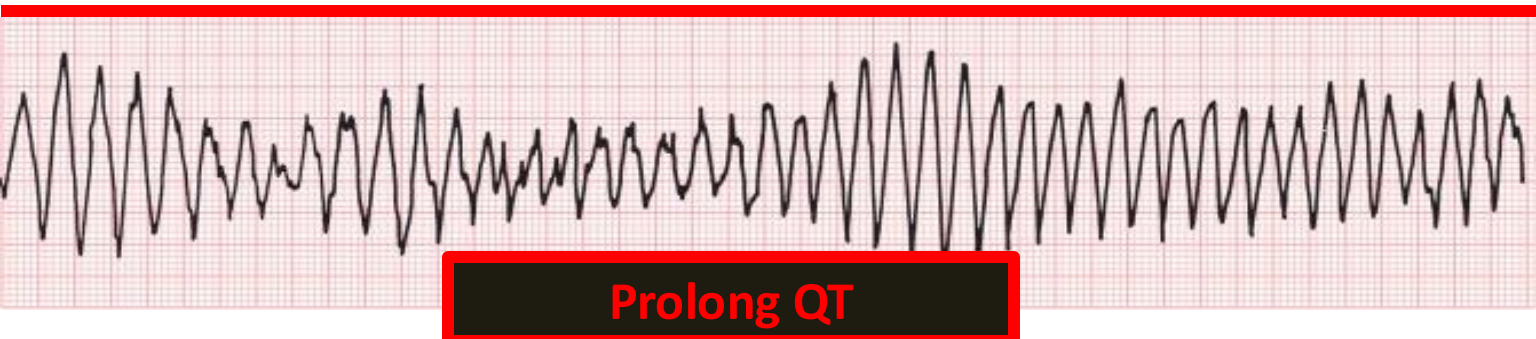
Ventricular tachycardia



Monomorphic VT



Polymorphic VT



Torsade de pointes

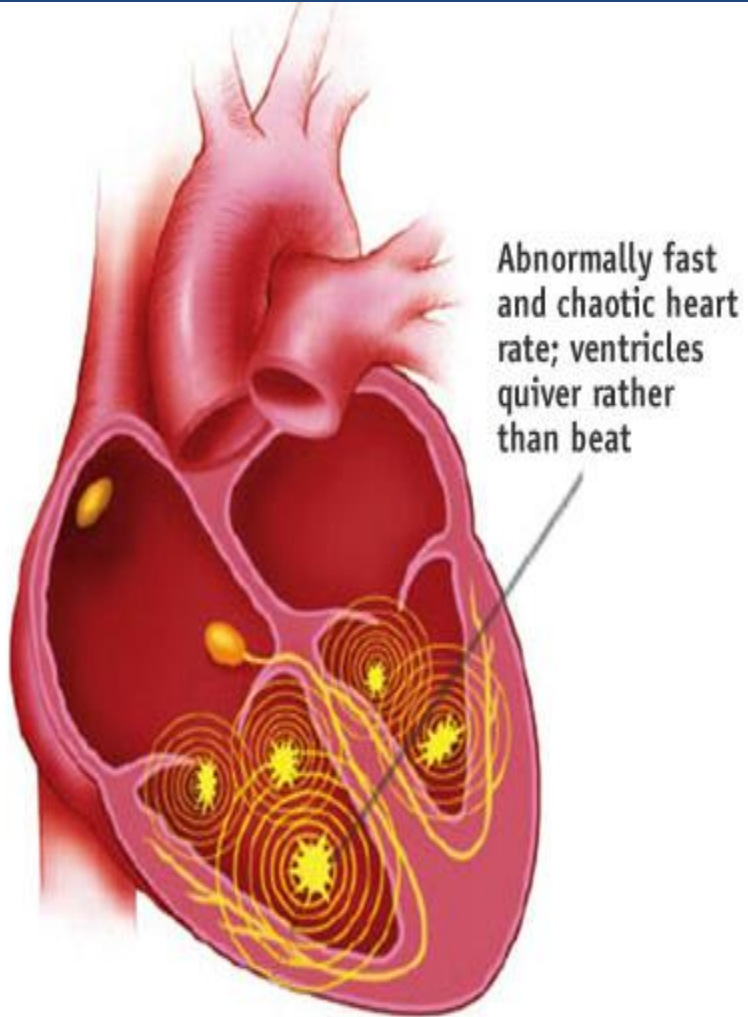


Ventricular tachycardia

Rhythm	Regular/ Irregular
Rate	150 -250 BPM. , > 250 BPM
P wave	-
PR interval	-
QRS	> 0.12 sec

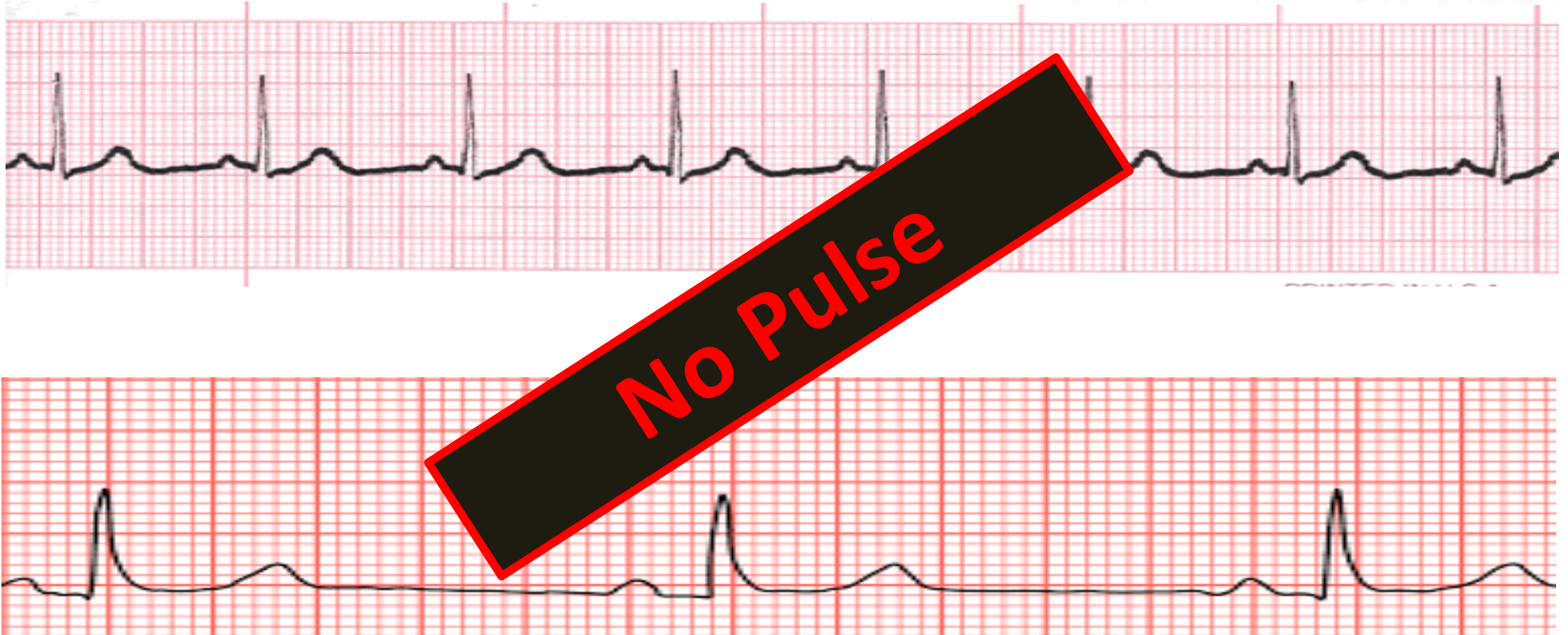


Ventricular fibrillation (VF)



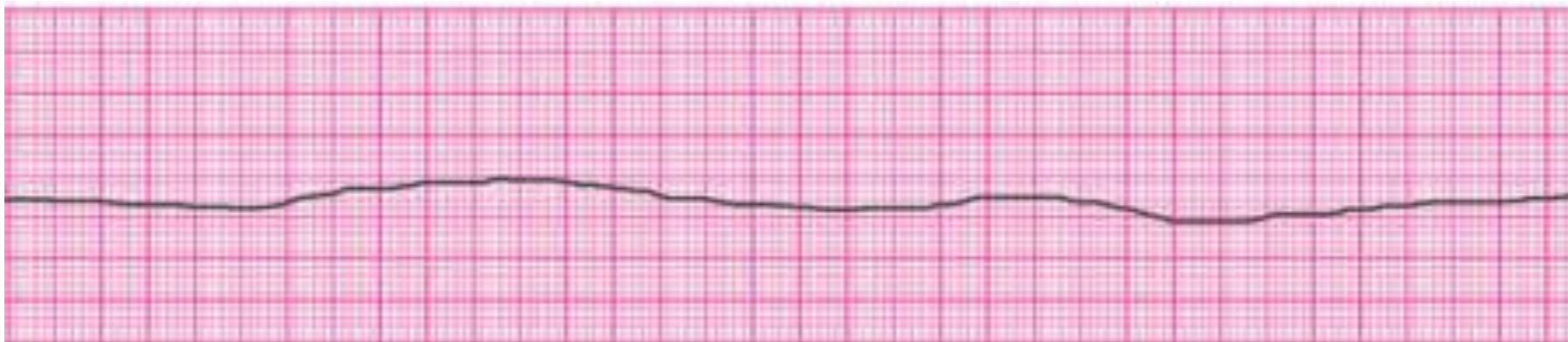


Pulseless Electrical Activity





Asystole



Size EKG

Check lead EKG

Common causes of tachyarrhythmia

Tachyarrhythmias	Cause
PAC & PVC	• Electrolyte imbalance; hypo K⁺ , hypo Mg⁺ • Heart; MI, CHF • Stress • Drugs; Digoxin • Caffeine, Alcohol
SVT	• Electrolyte abnormalities • Heart; MI, Cardiomyopathy, Enlargement, VHD • Drugs; Beta stimulation • Chronic lung disease (MAT), Hyperthyroidism • Digoxin toxicity, caffeine, and amphetamines

Common causes of tachyarrhythmia

Tachyarrhythmias	Cause
VT ,VF	• <u>Cardiac cause</u>; - Coronary heart disease; MI - VHD - Cardiomyopathy - Myocarditis • <u>Non Cardiac cause</u>; - Drug or toxicity → amiodarone, digitalis toxicity - Electrolyte disturbances ;hypocalcemia, hypokalemia and hypomagnesemia - Electrical shock, Drowning

Common causes of tachyarrhythmia

Arrhythmias	Cause
PEA & Asystole	• <u>5H</u> - Hypokalemia - Hypothermia - Hydrogen ion excess • <u>5T</u> - Tamponade (cardiac) - Thrombosis (MI) - Toxin
	- Hypoxia - Hypovolemia - Tension pneumothorax - Thrombosis (PE)



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MANAGEMENT



NCT

Stable

- Vagal maneuvers
- *If Regular rhythm → adenosine*
- Control rhythm or Ventricular rate ;
Beta –blocker or Calcium channel
blocker
- Treat underlying cause

Unstable

- *Synchronized cardioversion*
- If Regular rhythm → adenosine
- Treat underlying cause



WCT

Stable

- *If Regular wide monomorphic → adenosine*
- Antiarrhythmic infusion drip;
Amiodarone
- Treat underlying cause

Unstable

- Synchronized Cardioversion
- VF or pulseless VT
→ Defibrillation & CPR & Epinephrine & Amiodarone IV/IO
- Treat underlying cause



- Assessment
- Intervention
- Evaluation



- **Assessment**

- History; Underlying disease

- Physical ; V/S, stable??

- ❖ Hypotension

- ❖ Acute altered mental status

- ❖ Ongoing chest pain

- ❖ HF

- ❖ Signs of shock.



- Assessment
- Intervention
- Evaluation



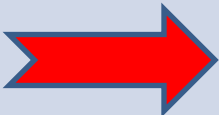
• **Intervention**

- Oxygen support
- Monitor EKG, EKG 12 leads
- IV Access → LAB; blood for electrolyte, Magnesium, Calcium, Cardiac enzymes, Thyroid functions
- Withholding or reducing the dose of positive chronotropes ; Dobutamine, Dopamine, Epinephrine
- Correct LAB or another cause



Treatment	Nursing Actions	Rationale
1. Non - Pharmacologic maneuvers - Vagal maneuvers; coughing, bearing down, breath holding, gag reflex carotid sinus massage	• V/S, Neuro signs • Monitor EKG • Prepare Emergency cart	• stimulate baroreceptors →CNS→acetylcholine→ slow down HR • assess carotid bruit → <u>Stroke</u>

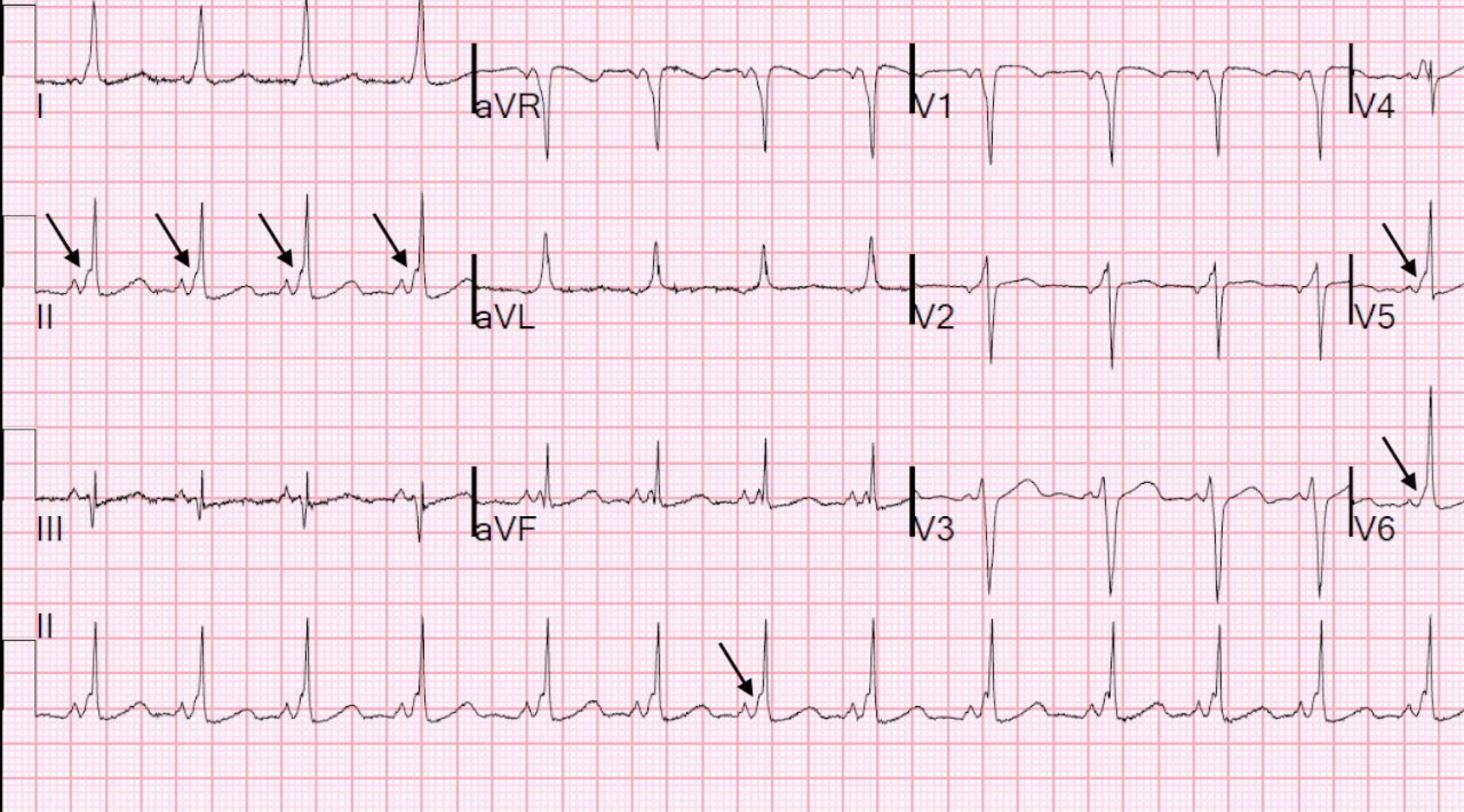


Treatment	Nursing Actions	Rationale
2. Pharmacologic 2.1 Adenosine	• Monitor EKG, V/S • Prepare Emergency cart  <u>Administration</u> • IV push over 1 to 2 seconds at a peripheral IV site at upper extremity 6→12→ 12mg. • Double syringe technique • Should not be used in patients with <u>Wolff-Parkinson-White (WPW) syndrome</u> → VF • Avoid use in patients with bronchoconstriction or bronchospasm	- Slows automaticity at the SA node, Slows conduction through the AV node -Half-Life: 10 sec -Side effect



Wolff-Parkinson-White (WPW)

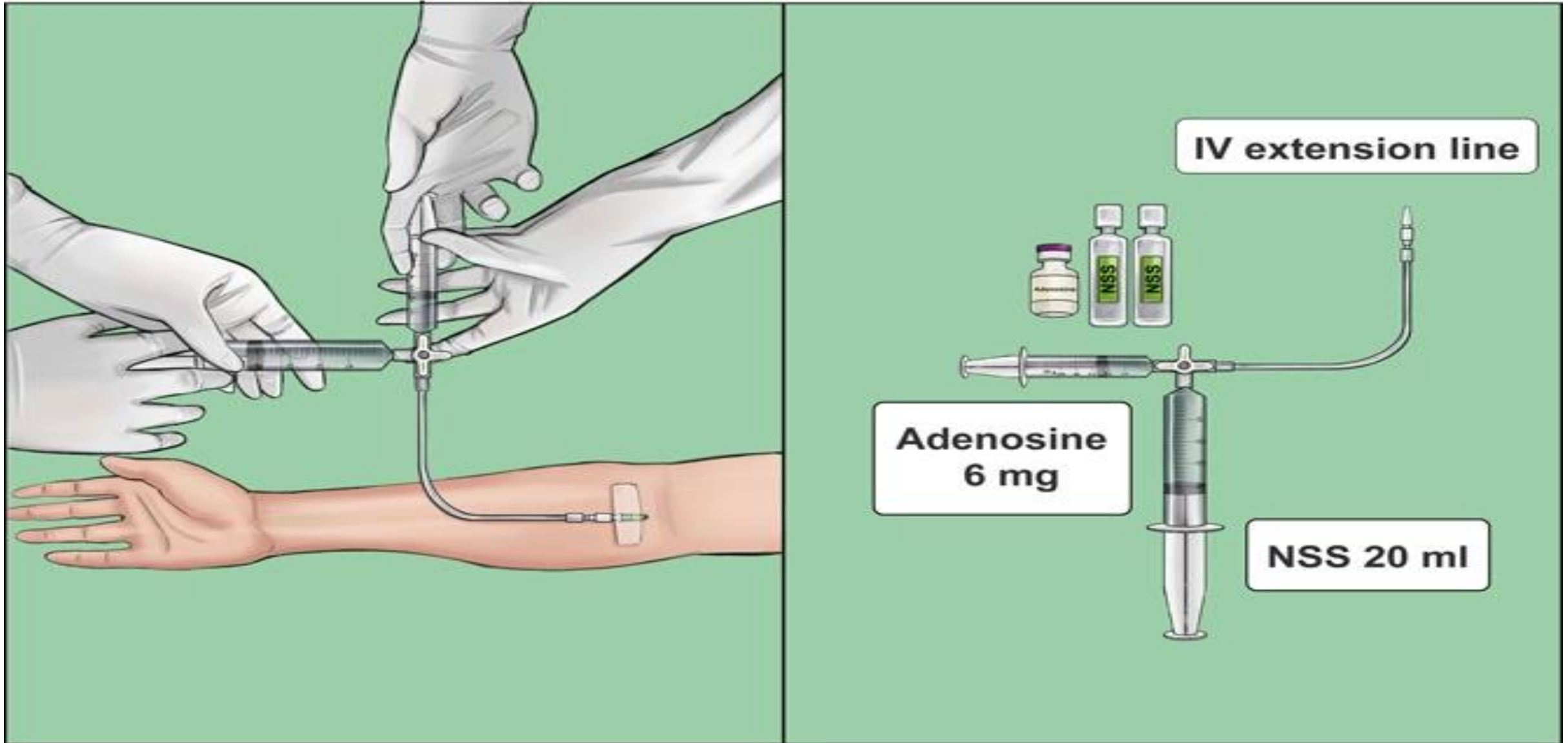






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Double syringe technique of Adenosine



[illegible]



Nursing

Treatment	Nursing Actions	Rationale
2. Pharmacologic 2.3 Amiodarone	• Monitor EKG, V/S • Prepare Emergency cart • F/U LFT (Hepatotoxicity) • Observe phlebitis	-Bradycardia -Hypotension -QT prolongation → Torsade de pointes





Treatment	Nursing Actions	Rationale
3. Electrical - Synchronized cardioversion - Defibrillation	• Inform the patient → Reason, may be discomfort • Analgesia or sedation • Monitor EKG, V/S, Neuro sign	



- Assessment
- Intervention
- Evaluation



- **Evaluation**

- Vital signs
- Symptoms
- Iatrogenic; Complication, Side effects



Conclusion

- Identify rhythm

- Stable or Unstable

- Management

- Non pharmacologic

- Pharmacologic

- Electrical

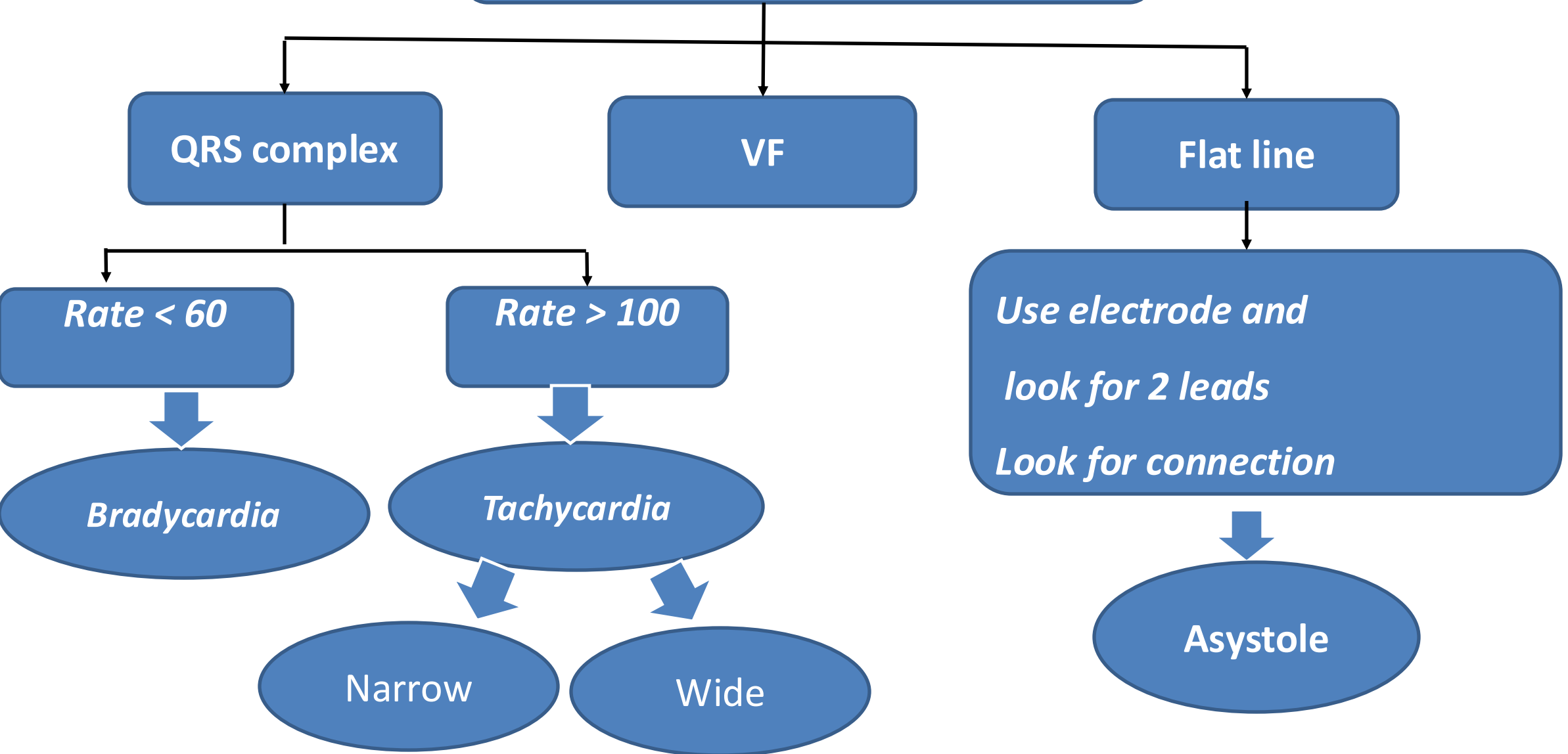


Correct cause

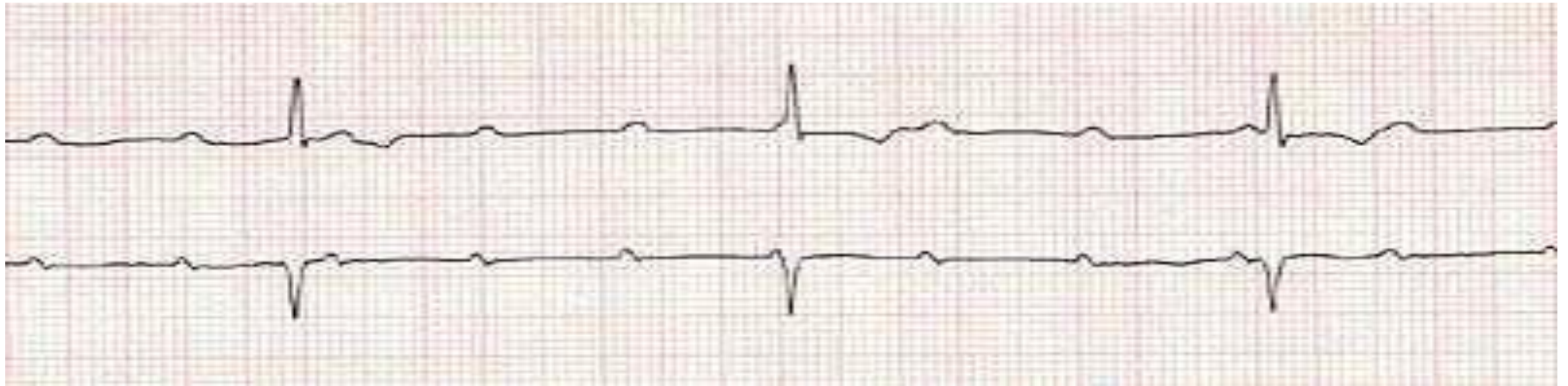
- Evaluation

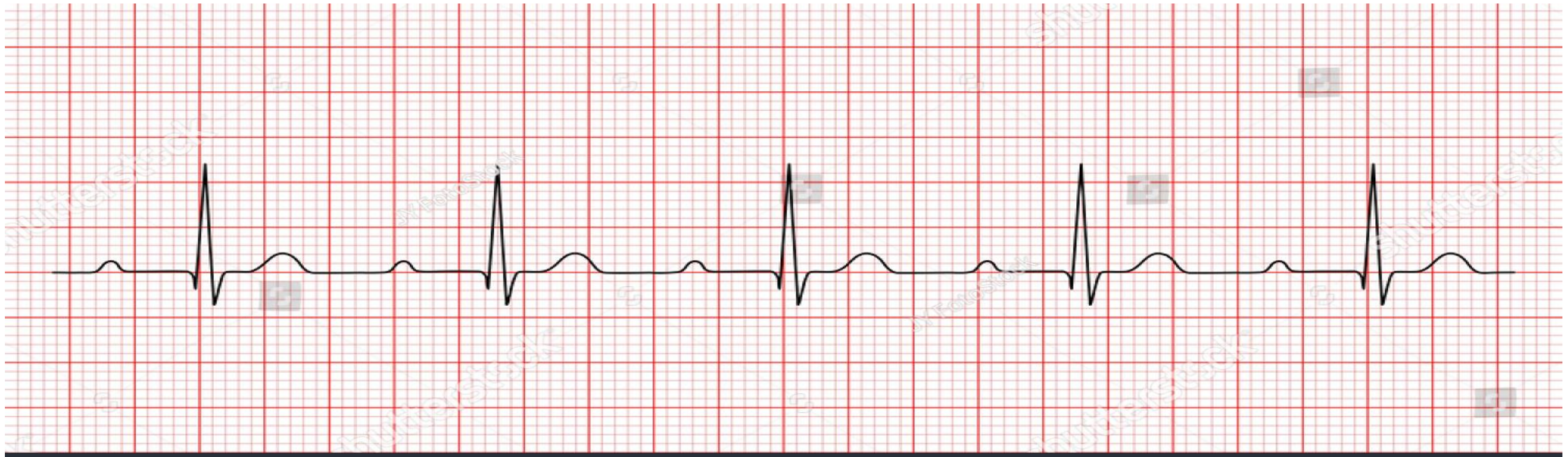


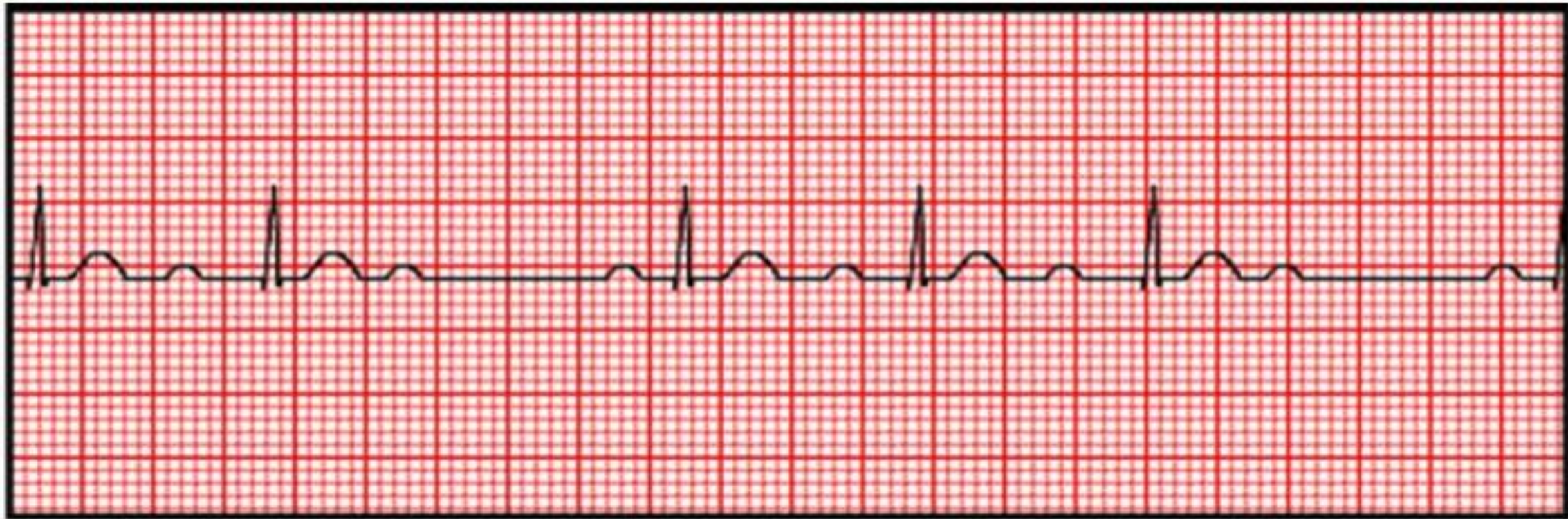
Rhythm

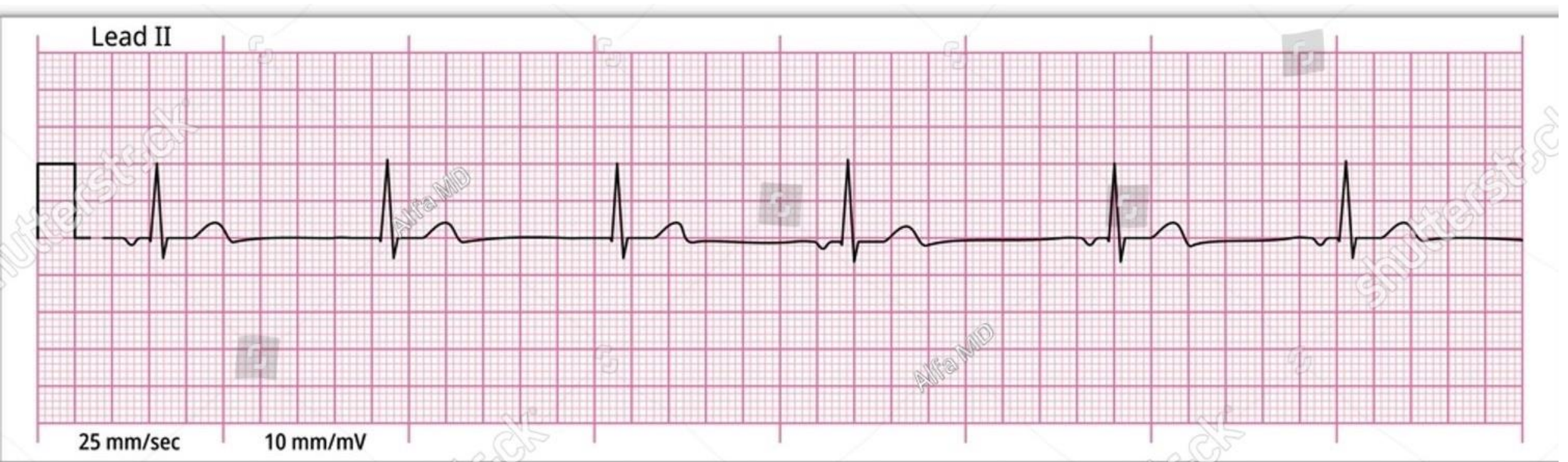


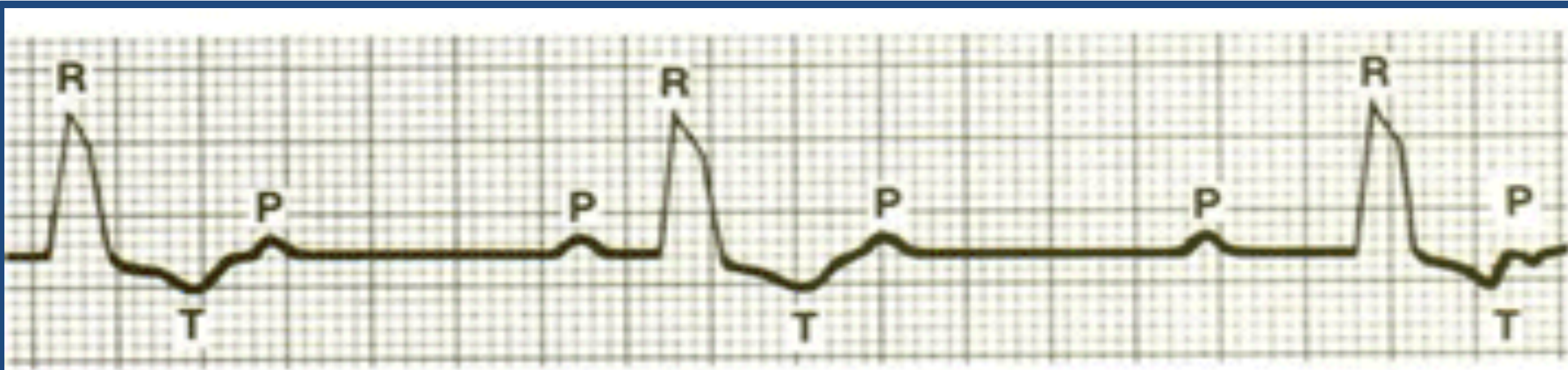
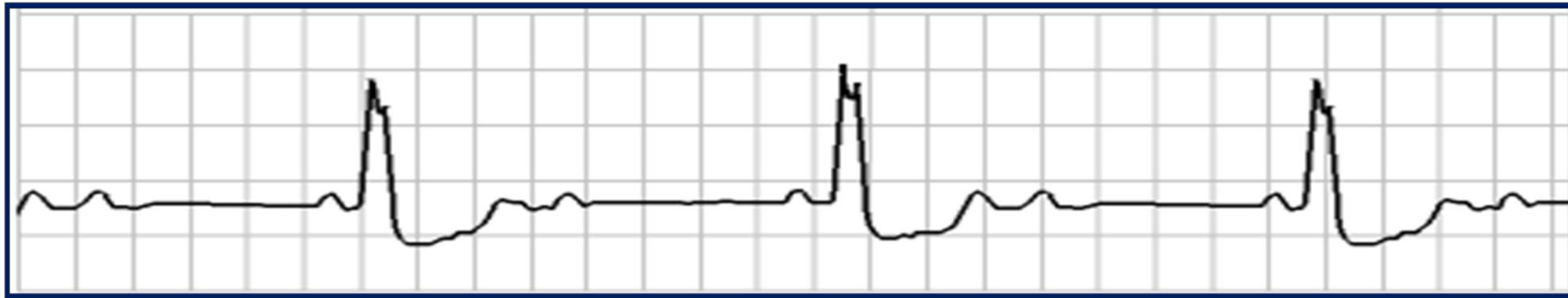














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Thank You